Algorithms and Computability

Lecture 7: NP-complete Problems

Martin Zimmermann (Aalborg University)

Last Week in Algorithms and Computability

We have seen:

- the complexity class NP: The class of languages accepted by polynomial-time NTM's.
- polynomial-time reductions.

Recall: $\mathrm{P} \subseteq \mathrm{NP}$.

The (literally) Million Dollar Question

Is verifying certificates easier than finding certificates or not: Is $\mathrm{P} = \mathrm{NP}$ or $\mathrm{P} \subsetneq \mathrm{NP}$?

- One of the deepest, most challenging, and most important questions of (theoretical) computer science.
- More on that next week.

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Polynomial-time Reductions

Definition

Let $A \subseteq \Sigma^*$ and $B \subseteq \Gamma^*$ be languages. We say that A is polynomial-time reducible to B, written $A \leq_P B$, if and only if

1. there is a polynomial-time computable function $f : \Sigma^* \to \Gamma^*$ such that
2. for every $w \in \Sigma^*$: $w \in A \Leftrightarrow f(w) \in B$.

Theorem

Let $A \leq_P B$.

1. *If $B \in \mathrm{P}$, then $A \in \mathrm{P}$.*
2. *If $B \in \mathrm{NP}$, then $A \in \mathrm{NP}$.*

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This result is used to prove **upper** bounds!

What About Lower Bounds?

Recall:

Theorem

Let $A \leq_m B$.

- *If A is not computable, then B is also not computable.*
- *If A is not computably-enumerable, then B is also not computably-enumerable.*

What About Lower Bounds?

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- Today, we will study how to obtain analogous results for P and NP.
- However, the fact that we have no problem that is proven to be in $\mathrm{NP} \setminus \mathrm{P}$ means we will identify candidates for such problems: the hardest problems in NP.
- **Caveat:** These hardest problem might still be in P, we just don't know.
- We will discuss this in more detail later today.

Agenda

1. Some More Problems
2. SAT Can Simulate All Problems in NP
3. NP-hardness and NP-completeness
4. More NP-complete Problems

Reminder: Graphs

We consider two types of graphs:

1. Directed Graphs: $G = (V, E)$ where
   - V is a finite set of vertices and
   - $E \subseteq V \times V$ is a set of (directed) edges.

Example

(V, E) with $V = \{1, 2, 3, 4\}$ and
$E = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$:

$$\begin{tikzcd} 4 \arrow[d] & 3 \arrow[l] \\ 1 \arrow[r] & 2 \arrow[u] \end{tikzcd}$$

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   - V is a finite set of vertices and
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2. Undirected Graphs: Directed graphs (V, E) such that $(v, v') \in E$ implies $(v', v) \in E$.

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Note that we drop the arrow tips in undirected graphs.

Reminder: The Clique Problem

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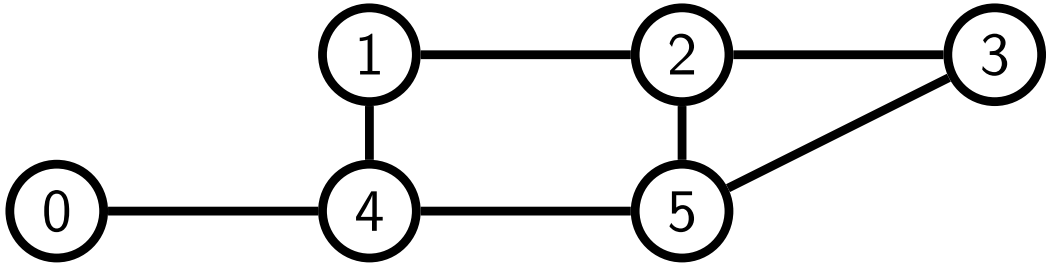

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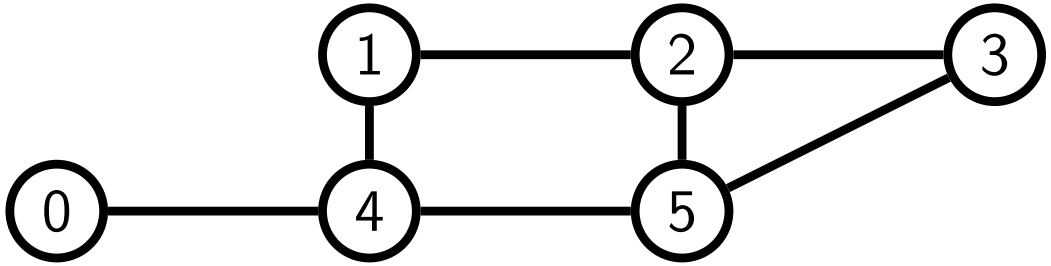

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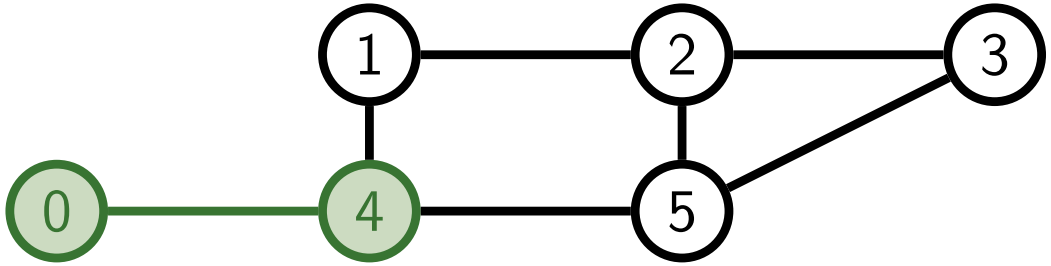

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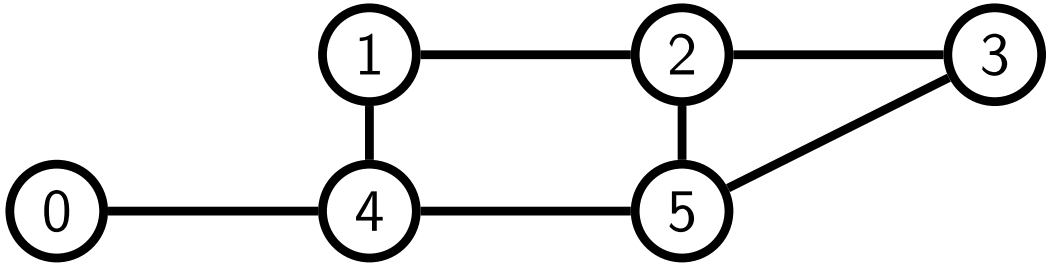

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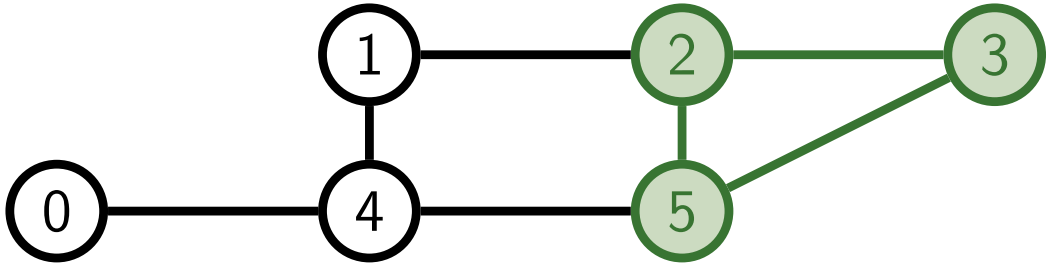

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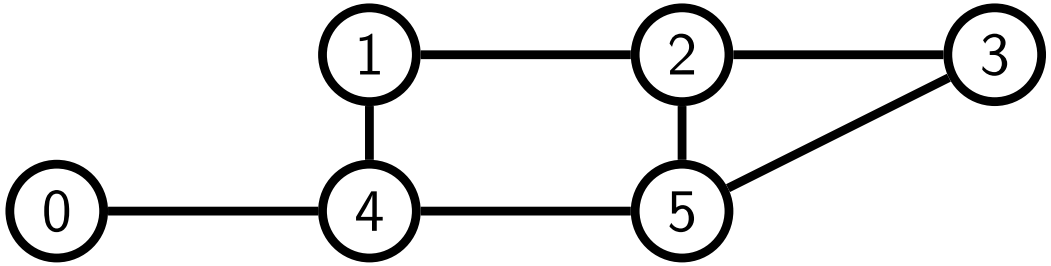

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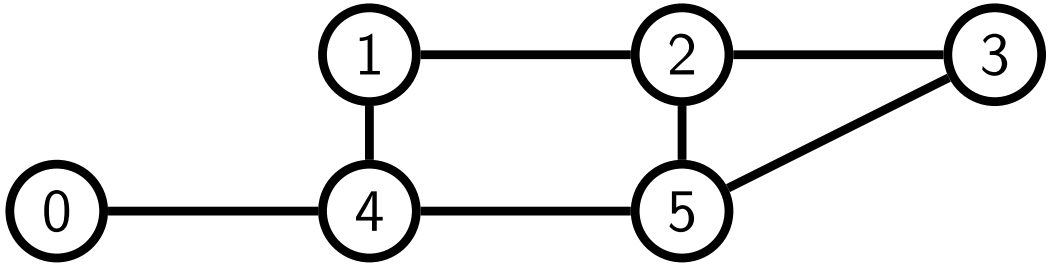

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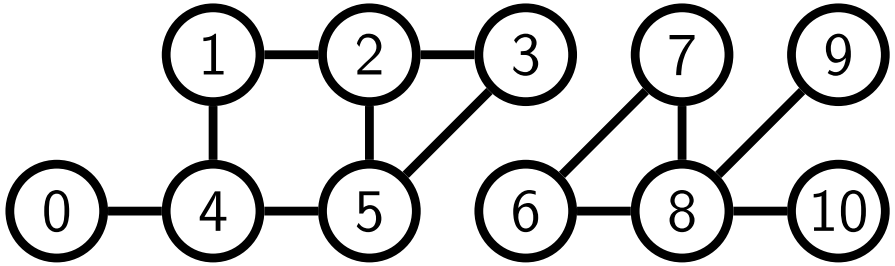

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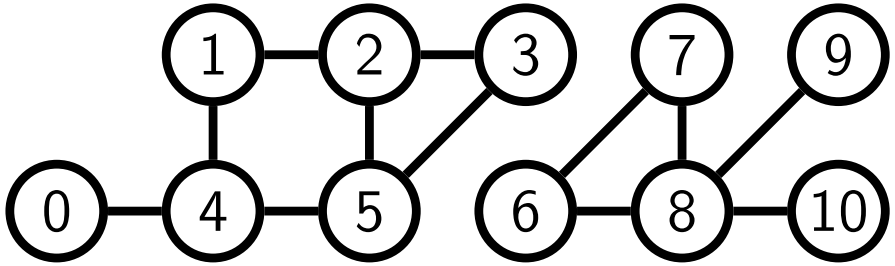

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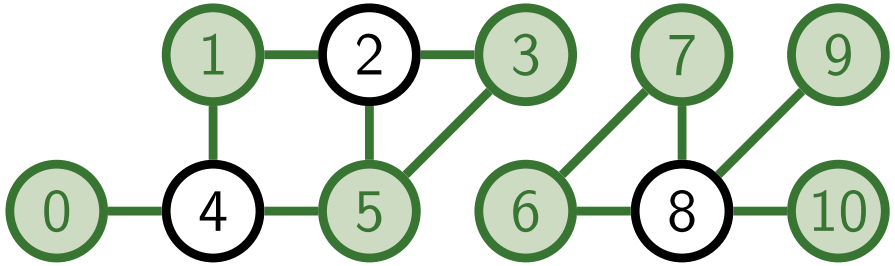

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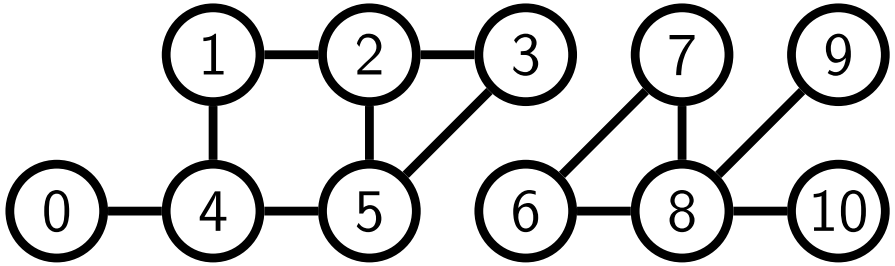

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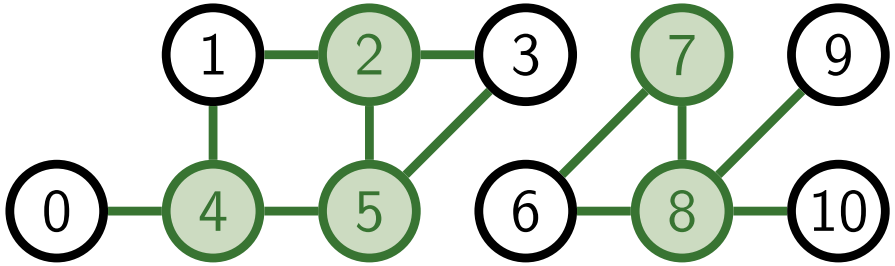

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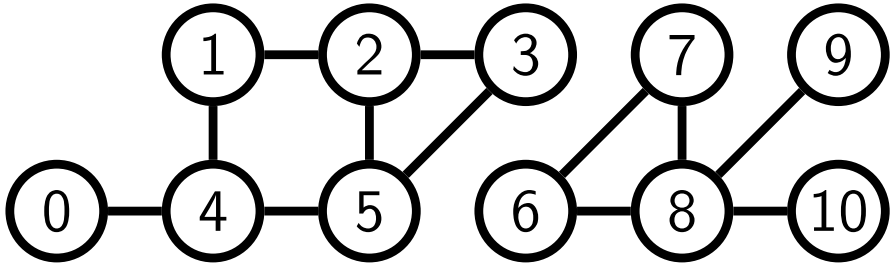

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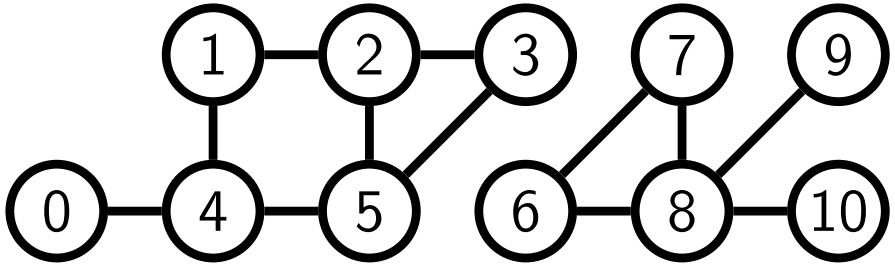

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- There are many other problems with the same status.
- Is there a reason we fail to show that these problems are in P?

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- There are many other problems with the same status.
- Is there a reason we fail to show that these problems are in P?

Yes, there is! They are some of the hardest problems in NP mentioned earlier.

Intuition

If only one of these hardest problems is in P, then $\mathrm{P} = \mathrm{NP}$ (i.e., all of them are)!

Agenda

1. Some More Problems
2. **SAT Can Simulate All Problems in NP**
3. NP-hardness and NP-completeness
4. More NP-complete Problems

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- Then, every branch of the computation tree induced by M and w has length at most $p(n)$.
- Also, M can visit at most $p(n) + 1$ cells before it terminates.

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- Also, M can visit at most $p(n) + 1$ cells before it terminates.
- For convenience, we encode a configuration $[q, \tau, \ell]$ by $\tau(1) \cdots \tau(\ell - 1)(\tau(\ell), q)\tau(\ell + 1) \cdots \tau(p(n) + 1) \in (\Gamma \cup (\Gamma \times Q))^{p(n)+1}$.

Setup

We want to show $L \leq_p \mathrm{SAT}$ for every $L \in \mathrm{NP}$.

- If $L \subseteq \Sigma^*$ is in NP, then there is a one-tape halting NTM M computing L with time-complexity $p(n)$ for some polynomial $p\colon \mathbb{N} \to \mathbb{R}_{>0}$.
- Fix some input $w = w_0 \cdots w_{n-1} \in \Sigma^*$ for M.
- Then, every branch of the computation tree induced by M and w has length at most $p(n)$.
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- Thus, a single branch of the tree can be represented by a $(p(n) + 1) \times p(n)$ grid such that entry at (i, j) represents the content of the i-th cell in the j-th configuration (and whether the head is at that cell, and the current state).

Visualization

$p(n)$	\$									\$
$\vdots$	\$									\$
	\$									\$
	\$									\$
	\$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	\$
	\$	X	1	$\#,q_\ell$	X	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
	\$	X	1	$\#$	$1,q_1^\#$	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
$\vdots$	\$	X	1	$\#,q_1$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
2	\$	X	$1,q_1$	$\#$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
1	\$	$1,s$	1	$\#$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
	0	1	$\cdots$					$\cdots$	$p(n)$	$p(n)+1$ $p(n)+2$

Visualization

$p(n)$	\$									\$
$\vdots$	\$									\$
	\$									\$
	\$									\$
	\$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	\$
	\$	X	1	$\#,q_\ell$	X	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
	\$	X	1	$\#$	$1,q_1^\#$	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
$\vdots$	\$	X	1	$\#,q_1$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
2	\$	X	$1,q_1$	$\#$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
1	\$	$1,s$	1	$\#$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
	0	1	$\cdots$					$\cdots$	$p(n)$	$p(n)+1$ $+2$

- Our goal: Given M and w compute in polynomial time a formula $\varphi_{M,w}$ such that $w \in L(M) \Leftrightarrow \varphi_{M,w} \in \mathrm{SAT}$.

Visualization

$p(n)$	\$									\$
$\vdots$	\$									\$
	\$									\$
	\$									\$
	\$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	\$
	\$	X	1	$\#,q_\ell$	X	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
	\$	X	1	$\#$	$1,q_1^\#$	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
$\vdots$	\$	X	1	$\#,q_1$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
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- Rely on locality: entry in (i,j) depends only on entries at $(i-1,j-1)$, $(i,j-1)$, and $(i+1,j-1)$: window of size 3×2.

Visualization

$p(n)$	\$									\$
$\vdots$	\$									\$
	\$									\$
	\$									\$
	\$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	\$
	\$	X	1	$\#,q_\ell$	X	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
	\$	X	1	$\#$	$1,q_1^\#$	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
$\vdots$	\$	X	1	$\#,q_1$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
2	\$	X	$1,q_1$	$\#$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	\$
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	0	1	$\cdots$					$\cdots$	$p(n)$	$p(n)+1$ $+2$

window(2,3)

X	1	$\#,q_1$
X	$1,q_1$	$\#$

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$\vdots$	$\$$									$\$$
	$\$$									$\$$
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$\vdots$	$\$$	X	1	$\#,q_1$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	$\$$
2	$\$$	X	$1,q_1$	$\#$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	$\$$
1	$\$$	$1,s$	1	$\#$	1	1	$\sqcup$	$\sqcup$	$\sqcup$	$\$$
	0	1	$\cdots$					$\cdots$	$p(n)$	$p(n)+1$ $+2$

window(2,3)

X	1	$\#,q_1$
X	$1,q_1$	$\#$

window(1,4)

$\$$	X	1
$\$$	X	1

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Roadmap

$$w \in L(M) \Leftrightarrow \varphi_{M,w} \in \mathrm{SAT}$$

1. $w \in L(M)$ is equivalent to the existence of a $(p(n) + 3) \times p(n)$-grid encoding an accepting run branch.
2. We express the existence of such a grid by a formula:

Roadmap

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1. $w \in L(M)$ is equivalent to the existence of a $(p(n)+3) \times p(n)$-grid encoding an accepting run branch.
2. We express the existence of such a grid by a formula:

$$\varphi_{M,w} = \varphi_{\text{setup}} \wedge \varphi_{\text{init}} \wedge \varphi_{\text{move}} \wedge \varphi_{\text{accept}}$$

using the propositions (for $i \in \{0, 1, \ldots, p(n)+2\}$, $j \in \{1, 2, \ldots, p(n)\}$, $a \in \Gamma \cup \{\$\}$, and $q \in Q$)

- $x_{i,j,a}$: grid entry (i, j) is a, and
- $x_{i,j,(a,q)}$: grid entry (i, j) is (a, q).

Setup

$$\varphi_{\text{setup}} = \bigwedge_{i=0}^{p(n)+2} \bigwedge_{j=1}^{p(n)} \bigvee_{e \in E} \left(x_{i,j,e} \wedge \bigwedge_{e' \in E \setminus \{e\}} \neg x_{i,j,e'} \right)$$

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Note

The size of φ_{setup} is $\mathcal{O}((p(n))^2 \cdot |E|^2)$ and therefore polynomial in $|\ulcorner M \urcorner| + n$.

Init

$$\varphi_{\text{init}} = x_{0,1,\$} \wedge x_{1,1,(w_0,s)} \wedge \left(\bigwedge_{i=2}^{n} x_{i,1,w_{i-1}} \right) \wedge \left(\bigwedge_{i=n+1}^{p(n)+1} x_{i,1,\sqcup} \right) \wedge x_{p(n)+2,1,\$}$$

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with

$$\psi_{i,j} = \bigvee_{\begin{array}{|c|c|c|}\hline e_3 & e_4 & e_5 \\ \hline e_0 & e_1 & e_2 \\ \hline \end{array}} x_{i-1,j-1,e_0} \wedge x_{i,j-1,e_1} \wedge x_{i+1,j-1,e_2} \wedge x_{i-1,j,e_3} \wedge x_{i,j,e_4} \wedge x_{i+1,j,e_5}$$

where the disjunction ranges over all legal windows.

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- The accepting state appears somewhere in the grid.

Note

The size of φ_{setup} is $\mathcal{O}((p(n))^2)$ and therefore polynomial in $|\ulcorner M \urcorner| + n$.

Wrapping Things Up

- The function mapping $\ulcorner M \urcorner$ and w to $\varphi_{M,w}$ is computable in polynomial time.
- We have $w \in L(M) \Leftrightarrow \varphi_{M,w} \in \mathrm{SAT}$:

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- Hence, $L \leq_p \mathrm{SAT}$.

Agenda

1. Some More Problems
2. SAT Can Simulate All Problems in NP
3. **NP-hardness and NP-completeness**
4. More NP-complete Problems

Hardness and Completeness

So, every language in NP can in polynomial time be reduced to SAT and SAT is in NP.

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Definition

Let L be a language.

1. L is NP-hard, if $L' \leq_p L$ for all $L' \in \mathrm{NP}$.
2. L is NP-complete, if L is NP-hard and in NP.

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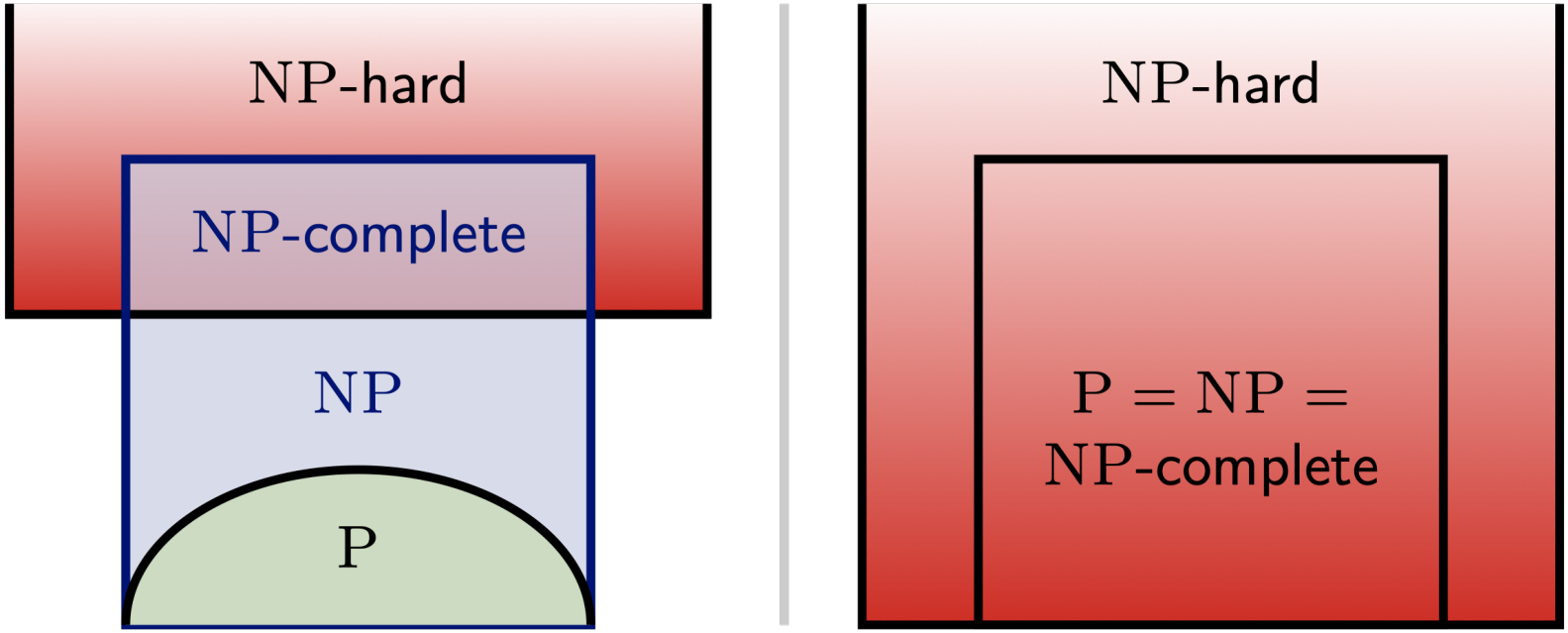
Corollary [Cook '61, Levin '63]

SAT is NP-complete.

Possible Worlds

We are in one of the following two worlds:



Why?

Theorem

Let L be NP-complete. If $L \in \mathrm{P}$, then $\mathrm{P} = \mathrm{NP}$.

P vs. NP

- In the 60 years since the Cook-Levin theorem, nobody has been able to show that an NP-complete problem is in P (and there are many candidates).
- Dually, nobody has shown that there is a problem in NP that is not in P.

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- Dually, nobody has shown that there is a problem in NP that is not in P.
- Most (but not all) computer scientists believe the latter to be the case.
- But this has been hard to prove! In fact, we even have theorems that say such a proof must be hard (in some very technical sense: there are barriers to separations), e.g., a diagonalization proof won't show $\mathrm{P} \neq \mathrm{NP}$.

The Exponential Time Hypothesis

Every halting DTM for SAT has exponential time-complexity.

Agenda

1. Some More Problems
2. SAT Can Simulate All Problems in NP
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4. **More NP-complete Problems**

CNF-SAT

A Boolean formula φ is in conjunctive normal form (CNF) if it is of the form

$$\varphi = \bigwedge_{i=1}^{n} \bigvee_{j=1}^{m_i} \ell_{i,j}$$

such that each $\ell_{i,j}$ is a literal, i.e., either a variable x or a negated variable $\neg x$.

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Decision problem:

$$\text{CNF-SAT} = \{\varphi \mid \varphi \text{ is in CNF and has a satisfying assignment}\}$$

CNF-SAT is NP-complete

Recall from the last exercise sheet: $\mathrm{DNF\text{-}SAT}$ is in P.

Theorem

$\mathrm{CNF\text{-}SAT}$ *is* NP*-complete.*

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$\varphi_{M,w}$ can easily be transformed in polynomial time to be in CNF (see exercises).

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Note

- Every Boolean Formula can be turned into an equivalent one in DNF and into an equivalent one in CNF.
- Thus, the structure of the formula has direct consequences for the complexity of the satisfiability problem!

How to Prove Problems NP-hard?

How can we prove that some L is NP-hard?

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Proof.

Show that $\leq_p$ is transitive, i.e., $L_1 \leq_p L_2$ and $L_2 \leq_p L_3$ implies $L_1 \leq_p L_3$. $\square$

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A Boolean formula φ is in 3CNF if it is of the form

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$(x \vee \neg y \vee z) \wedge (\neg x \vee \neg x \vee \neg x) \wedge (\neg y \vee \neg z \vee \neg z)$ is in 3CNF.

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Proof

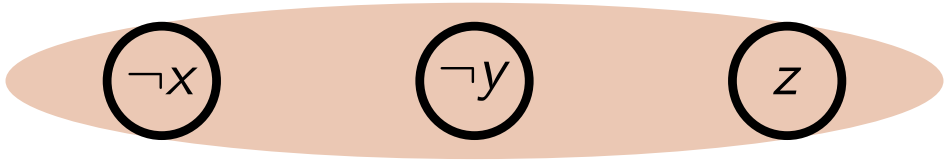
See the exercises.

Clique is NP-complete

- We show $3\mathrm{SAT} \leq_p \mathrm{CLIQUE}$..
- by illustrating the reduction on an example first.

Clique is NP-complete

Let $\varphi = (\neg x \vee \neg y \vee z) \wedge (x \vee y \vee \neg z) \wedge (x \vee y \vee z) \wedge (\neg x \vee y \vee \neg z)$



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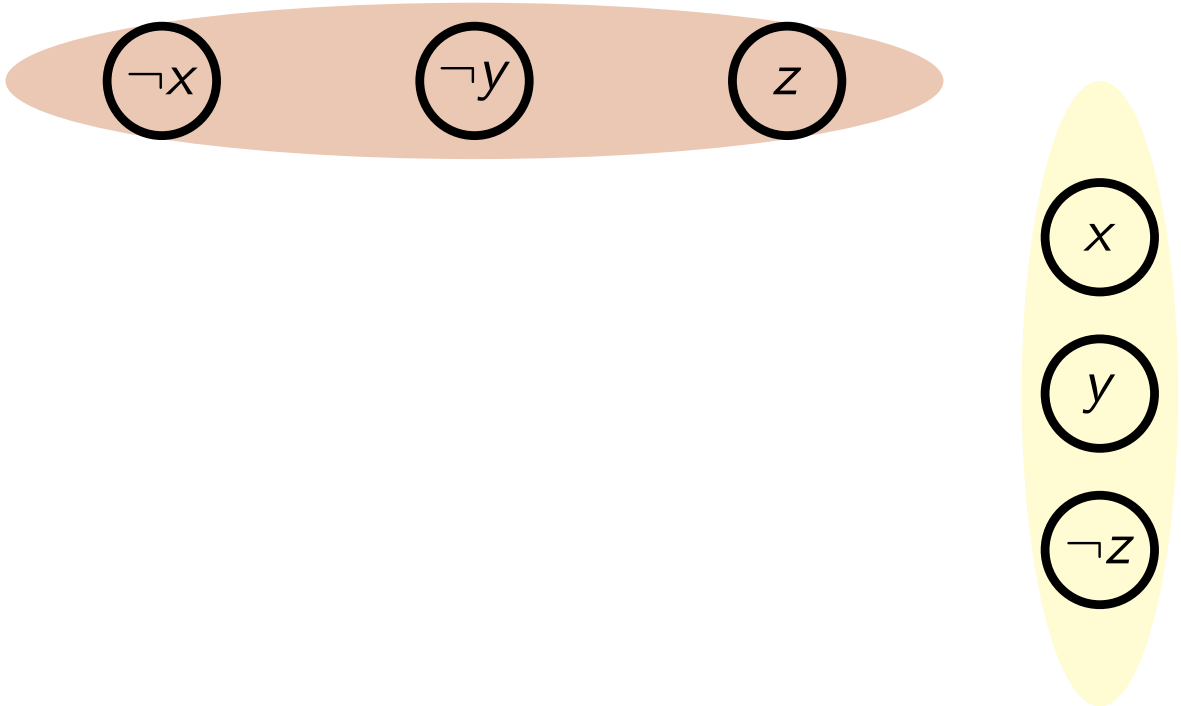

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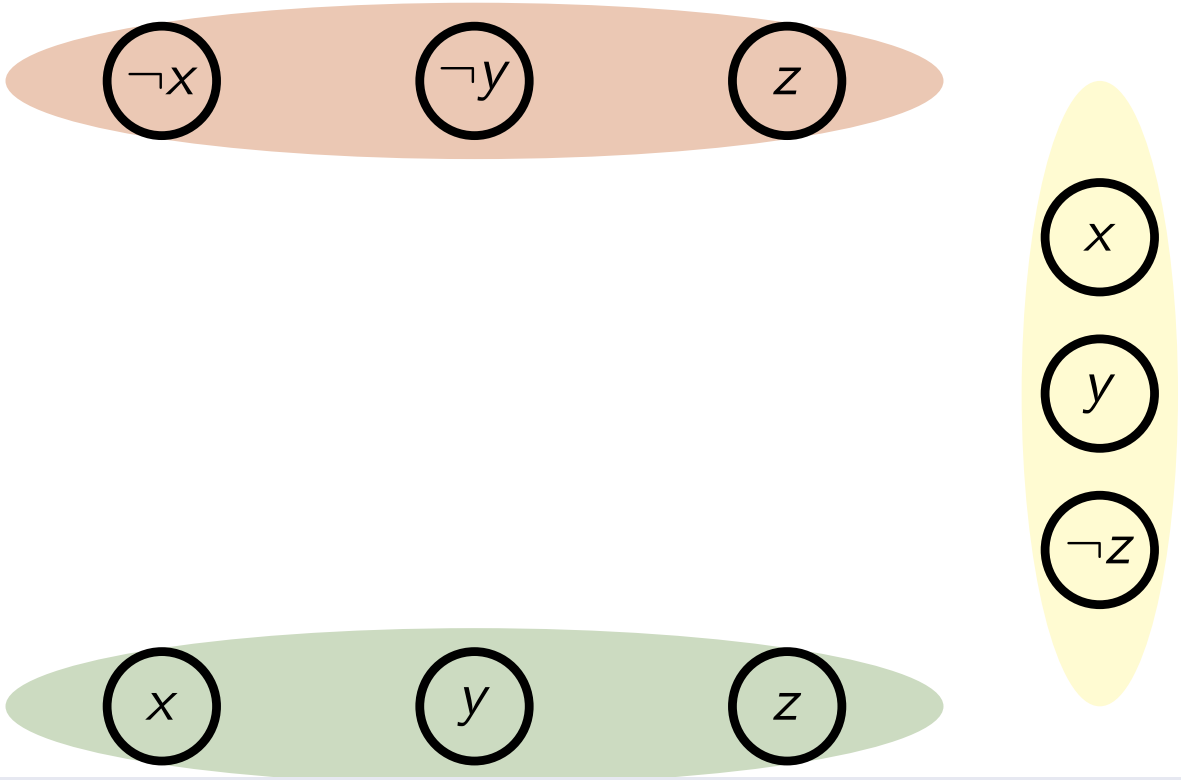

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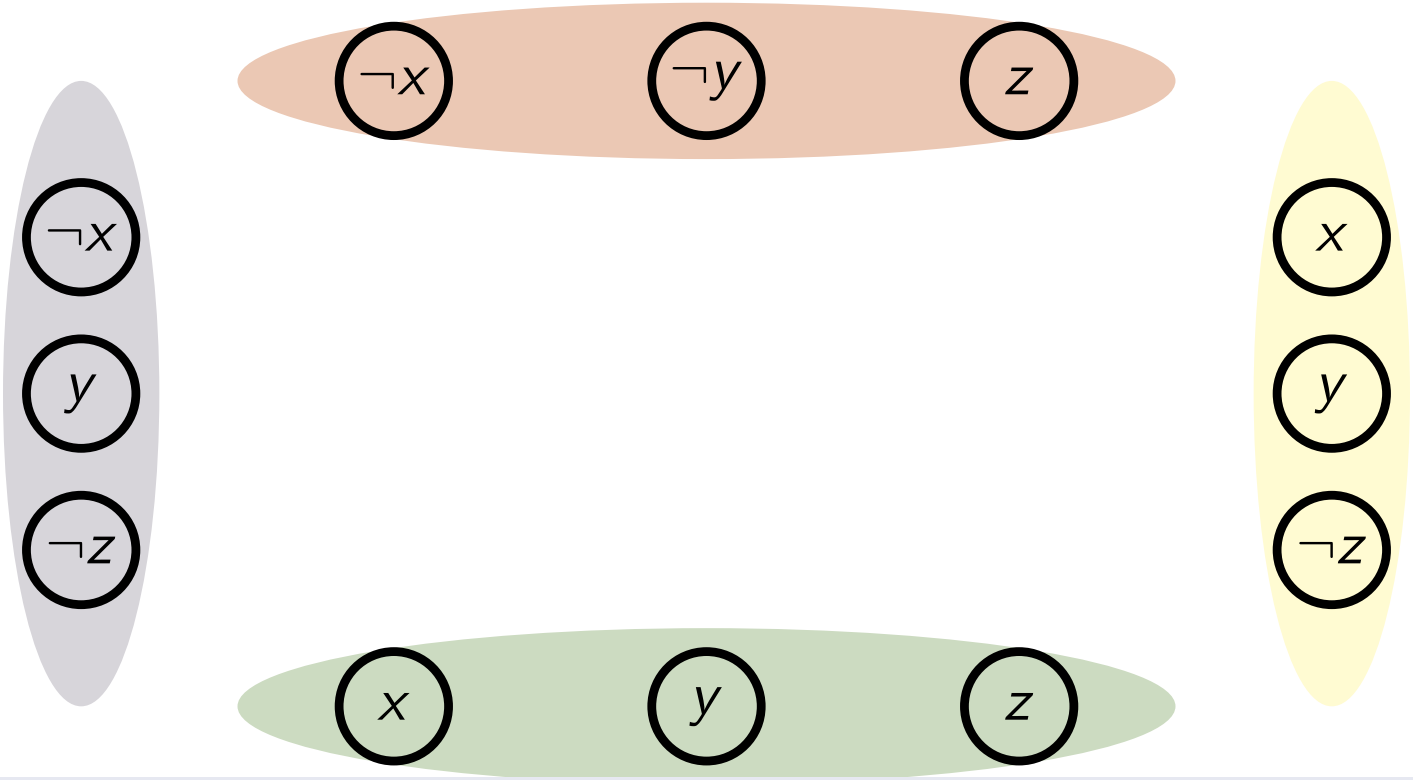

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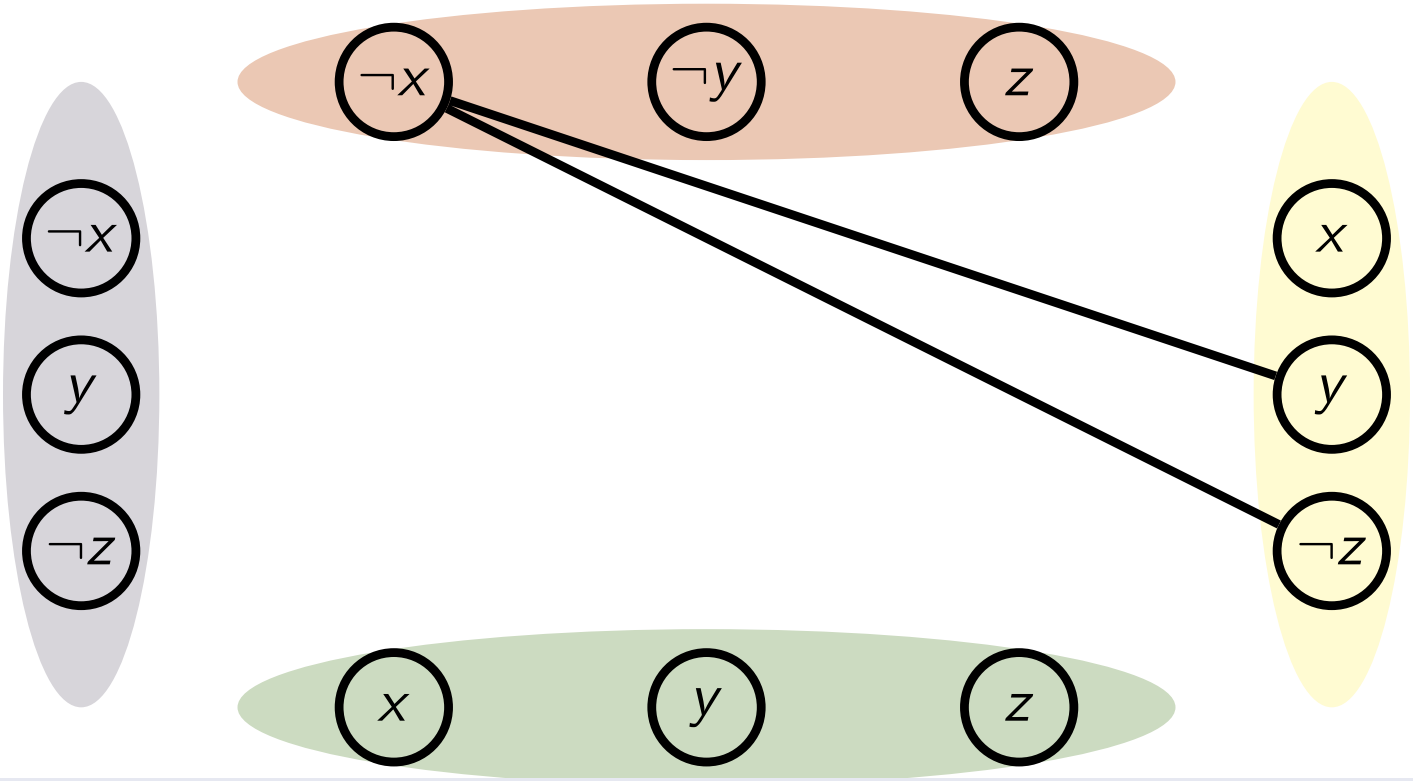

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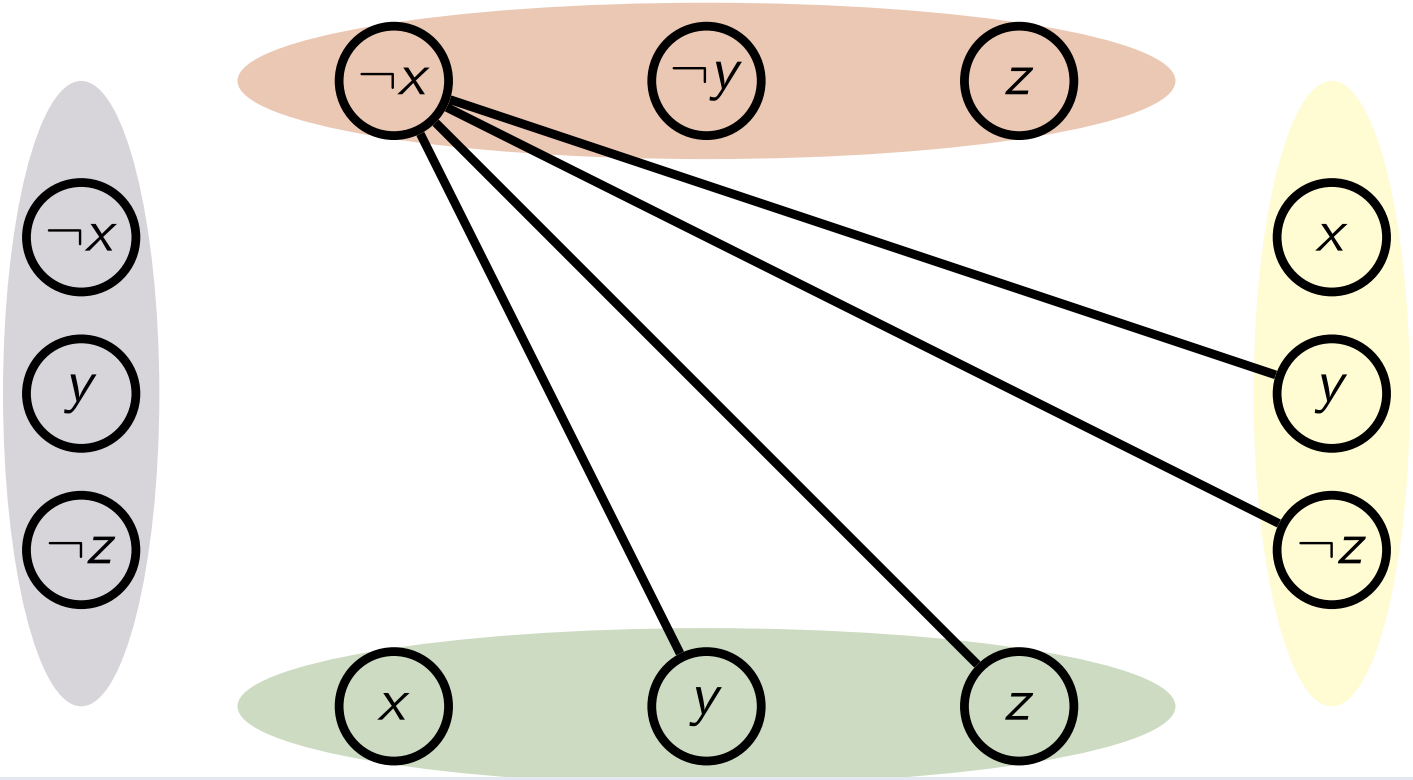

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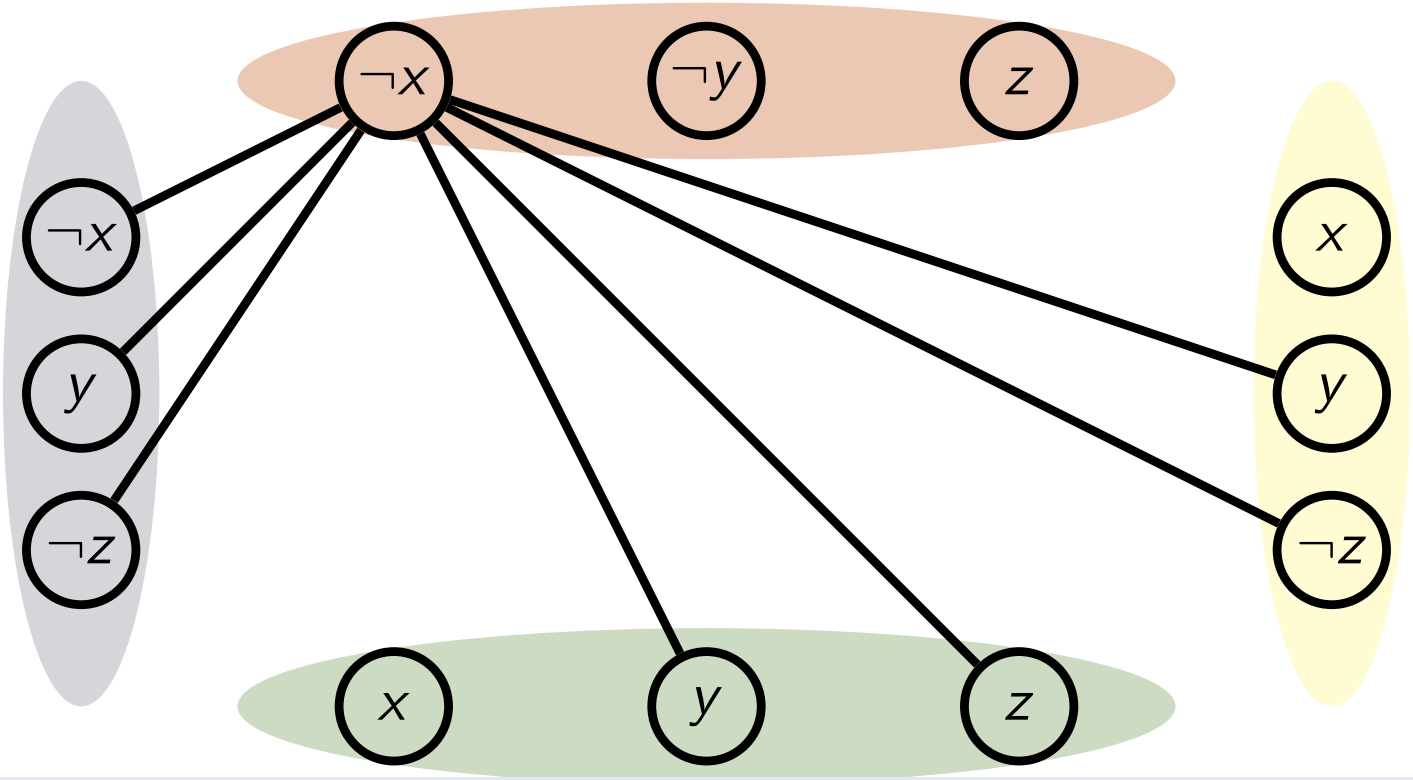

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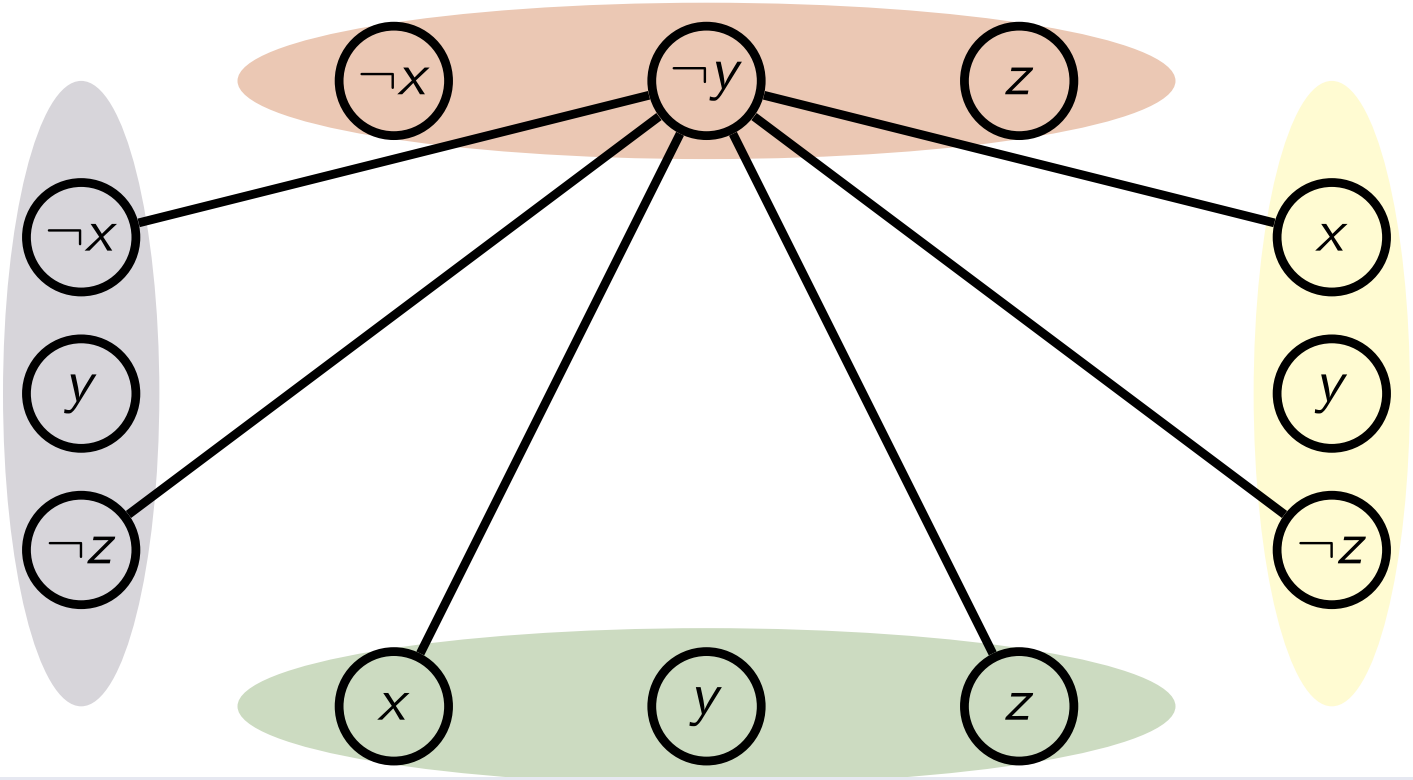

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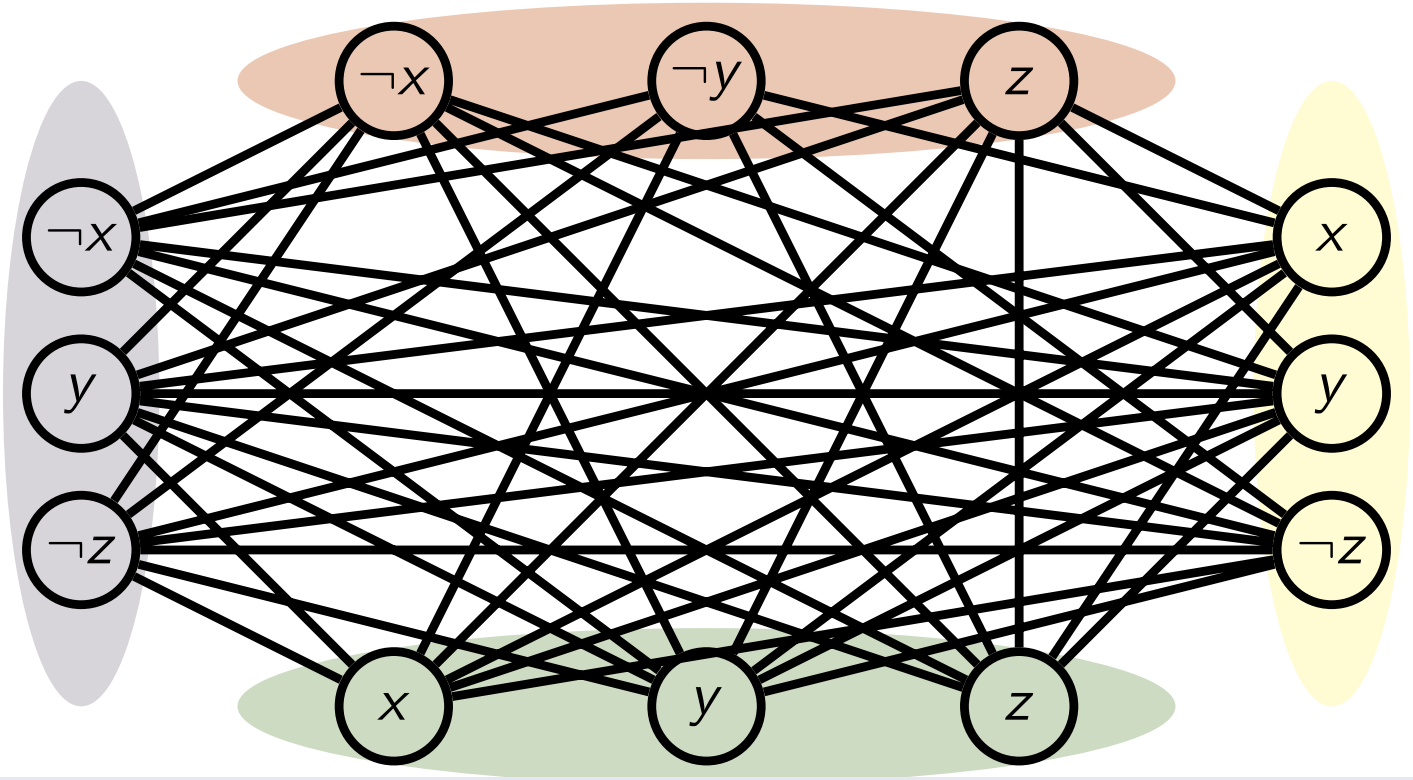

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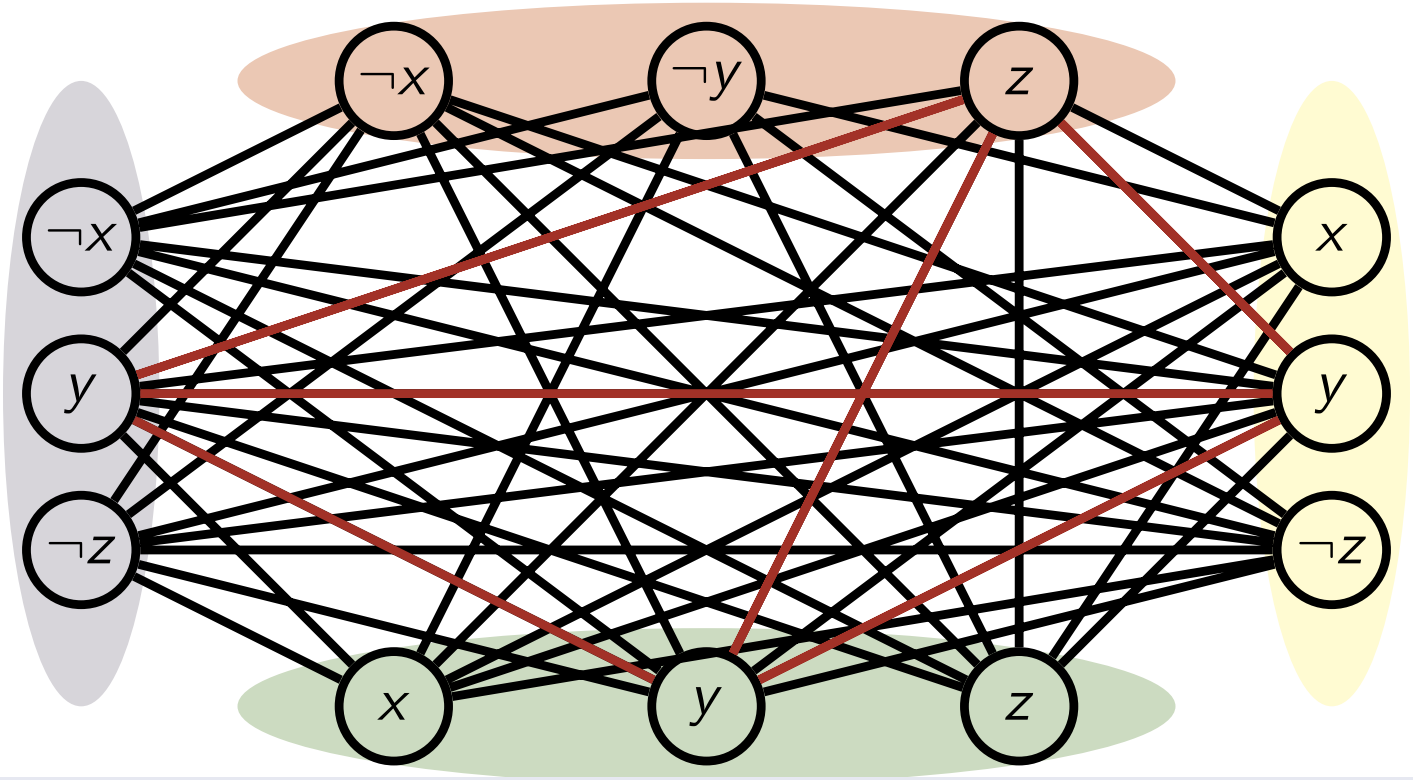
- A 4-clique contains exactly one vertex from each of the four clauses and no contradictory literals.
- This can be extended to a satisfying assignment.



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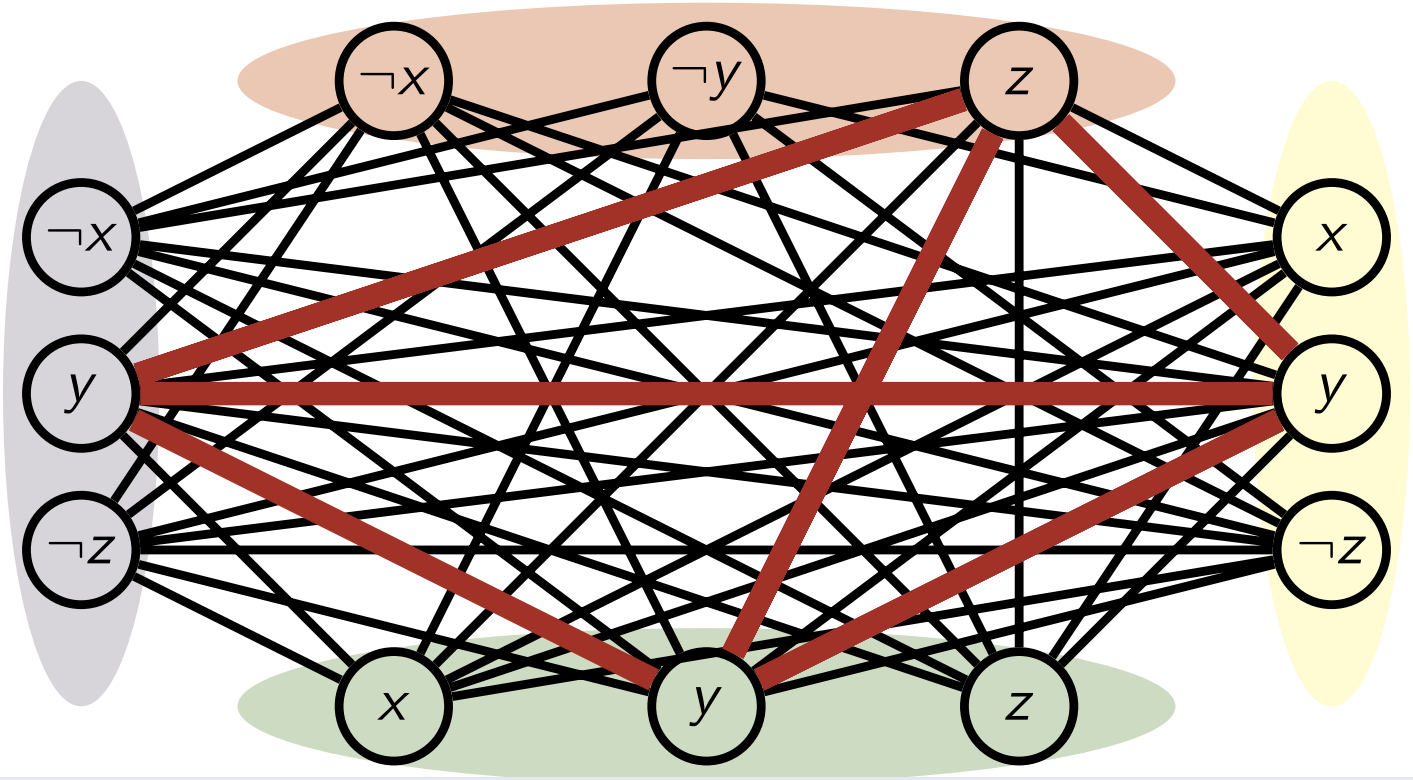

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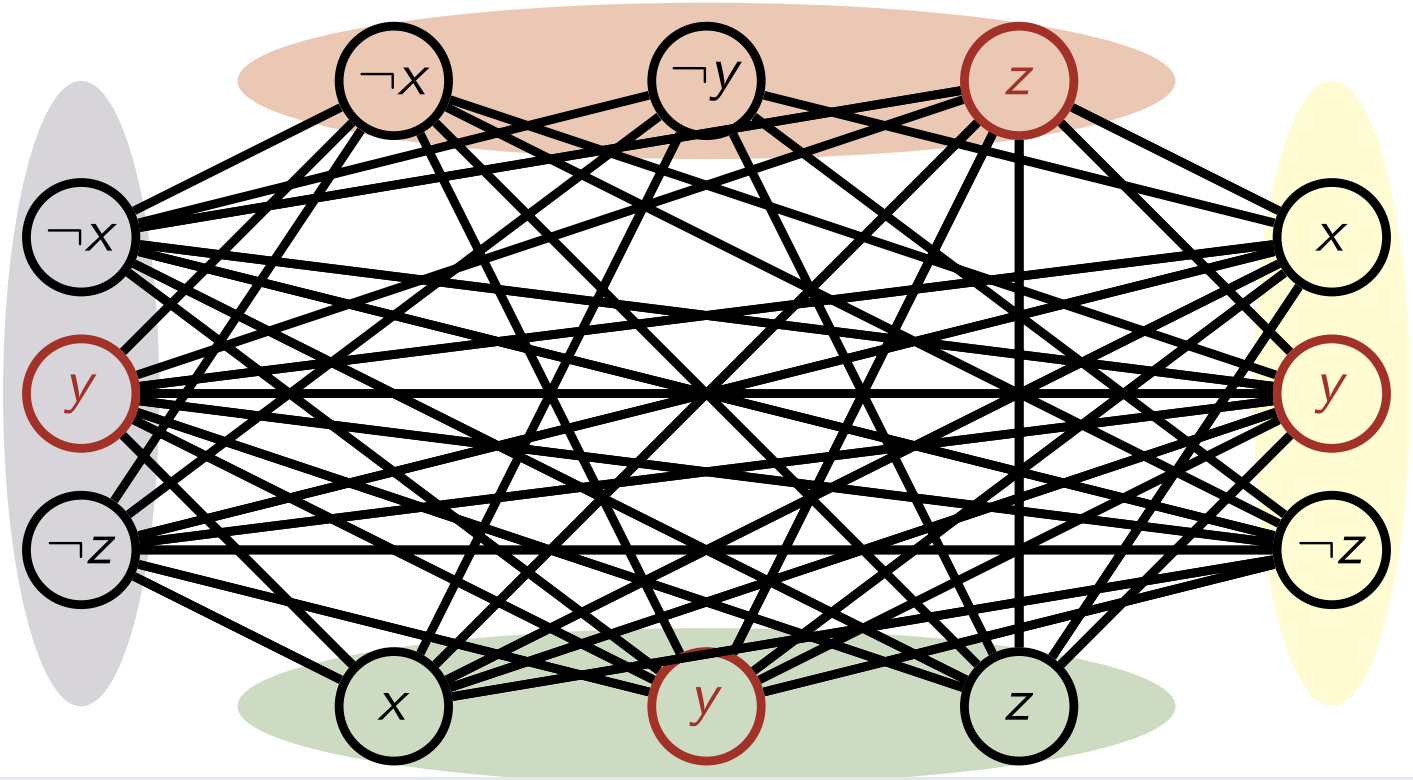
- Vice versa: A satisfying assignment satisfies some literal in each clause.
- Picking one from each clause gives a 4-clique.



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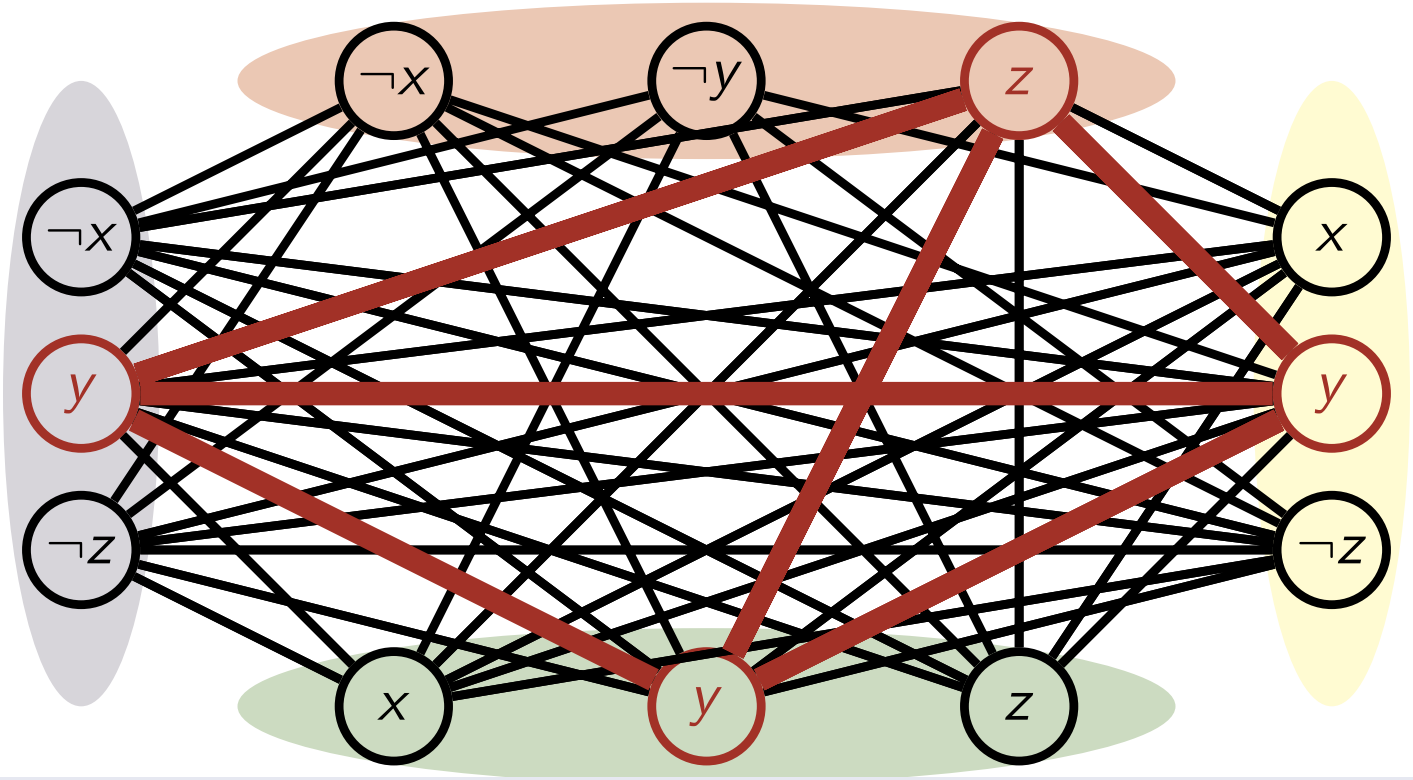

Clique is NP-complete

General construction for $\bigwedge_{i=1}^{n} \bigvee_{j=1}^{3} \ell_{i,j}$: Construct G_φ with

- One vertex for each literal $\ell_{i,j}$ ($3n$ nodes) and
- all edges but those between
  - literals in the same clause and
  - contradictory literals (e.g., x and $\neg x$).
- Then, we have $\varphi \in \mathrm{SAT} \Leftrightarrow (G_\varphi, n) \in \mathrm{CLIQUE}$.

Vertex Cover is NP-complete

- We show $3\mathrm{SAT} \leq_p \mathrm{VC}$..
- by illustrating the reduction on an example only.

Vertex Cover is NP-complete

Let $\varphi = (\neg x \vee \neg y \vee z) \wedge (x \vee y \vee \neg z) \wedge (x \vee y \vee z)$ with $v = 3$ variables and $c = 3$ clauses.

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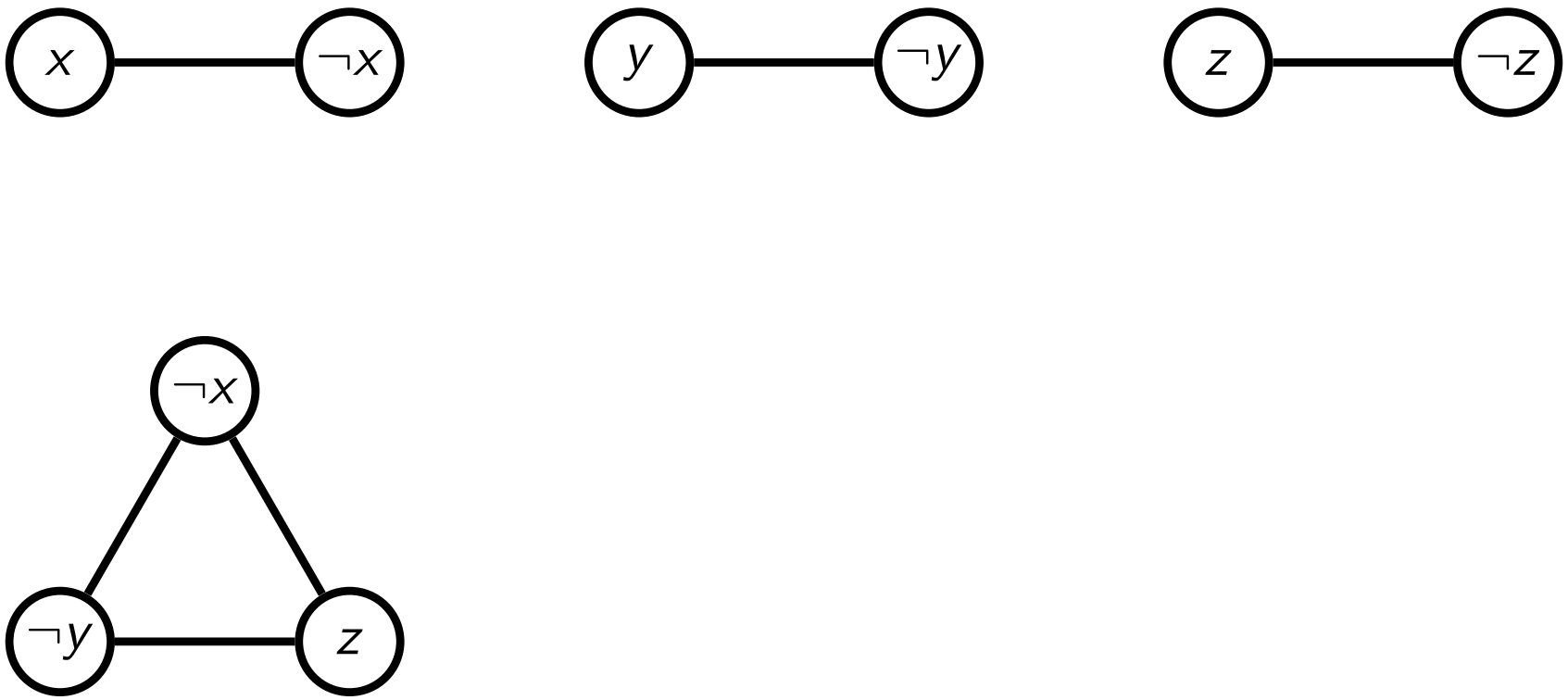

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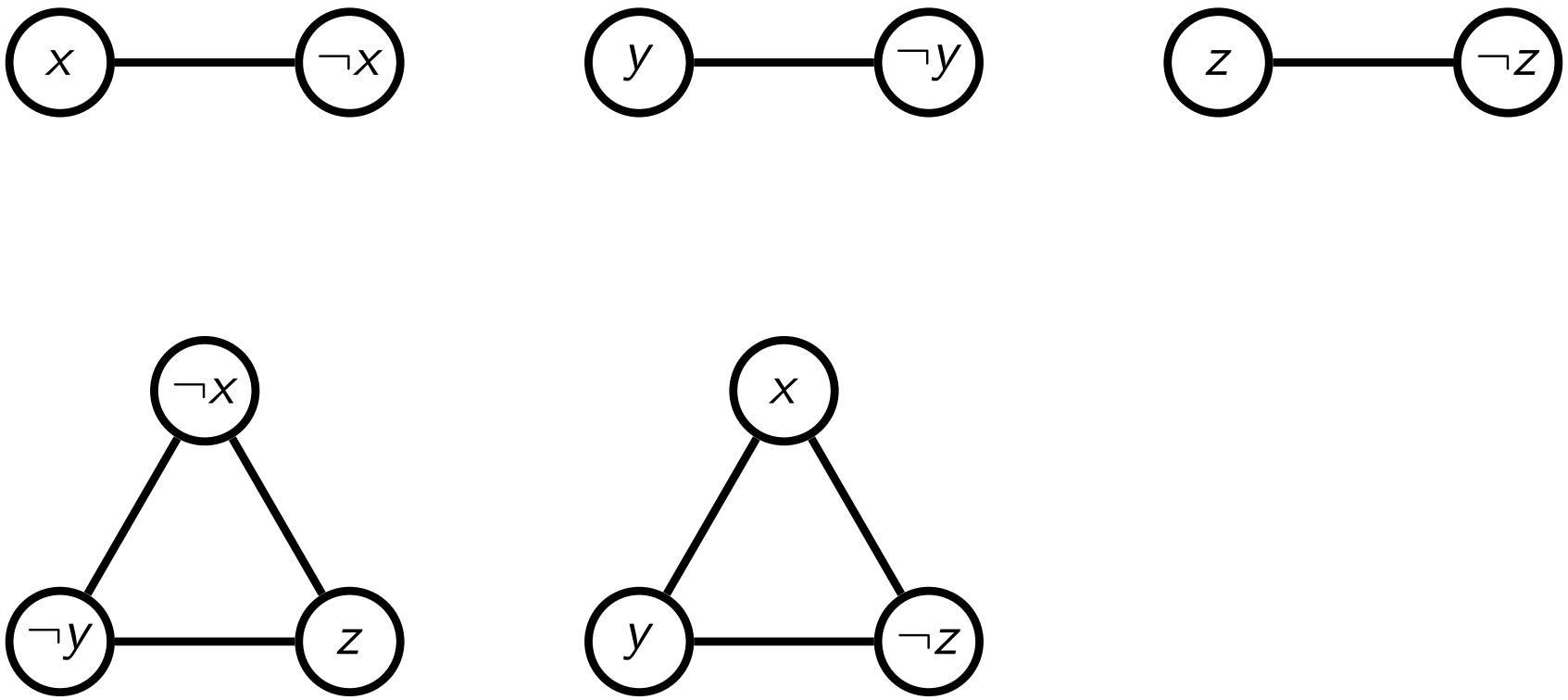

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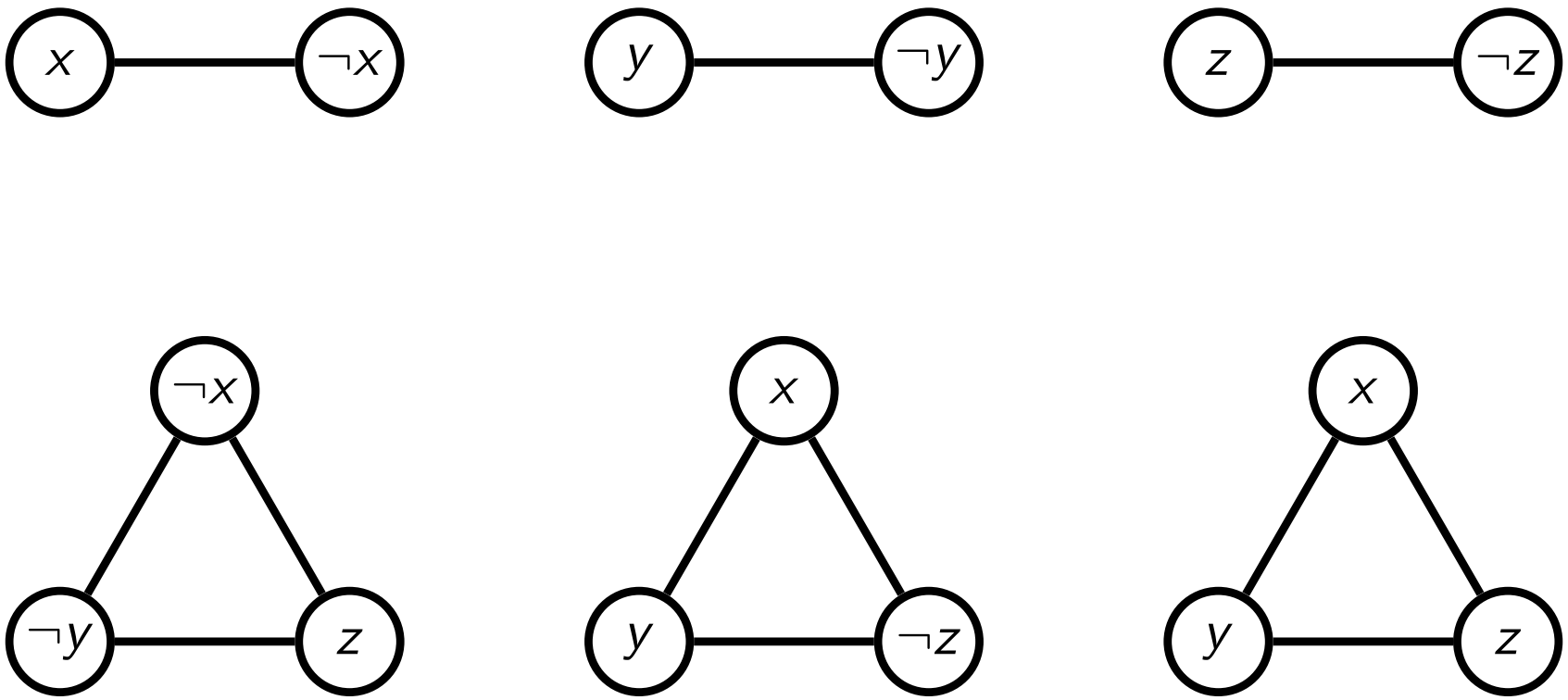

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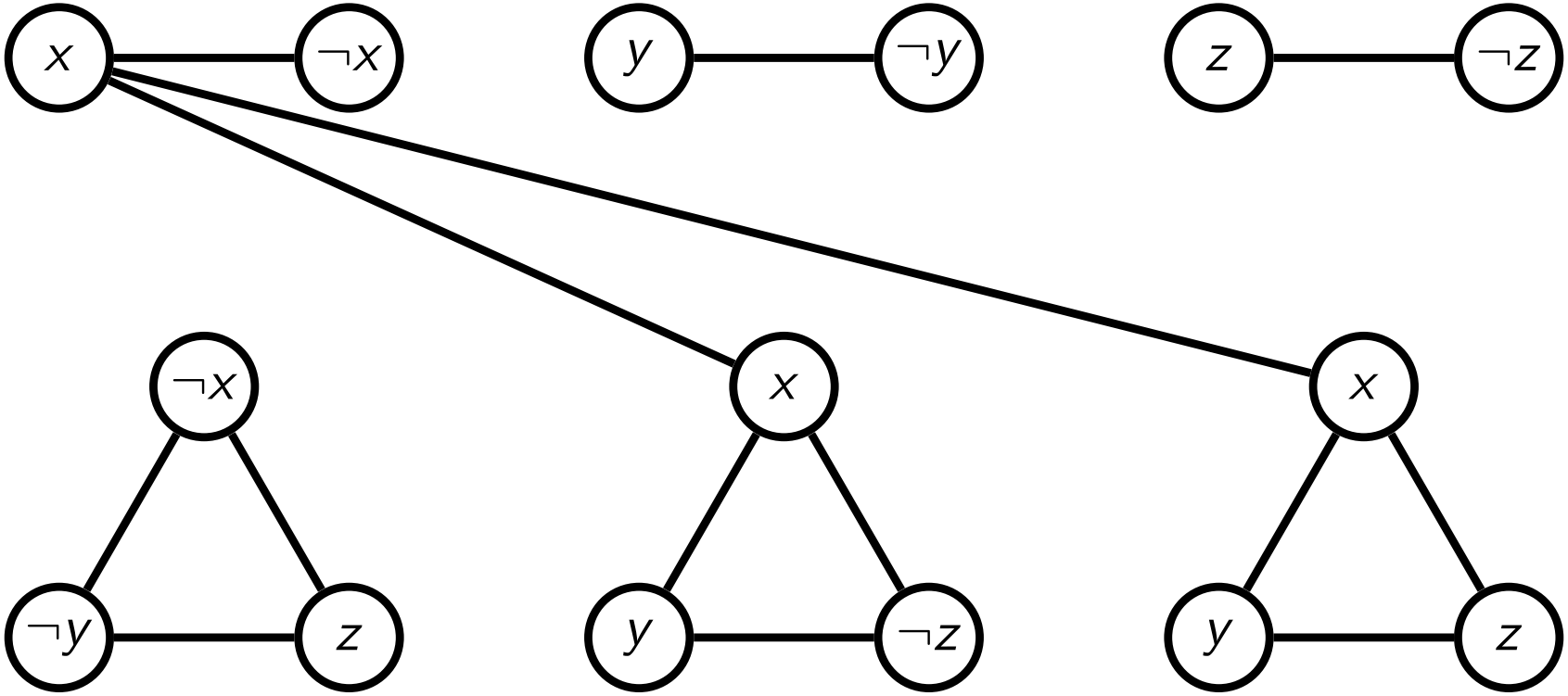

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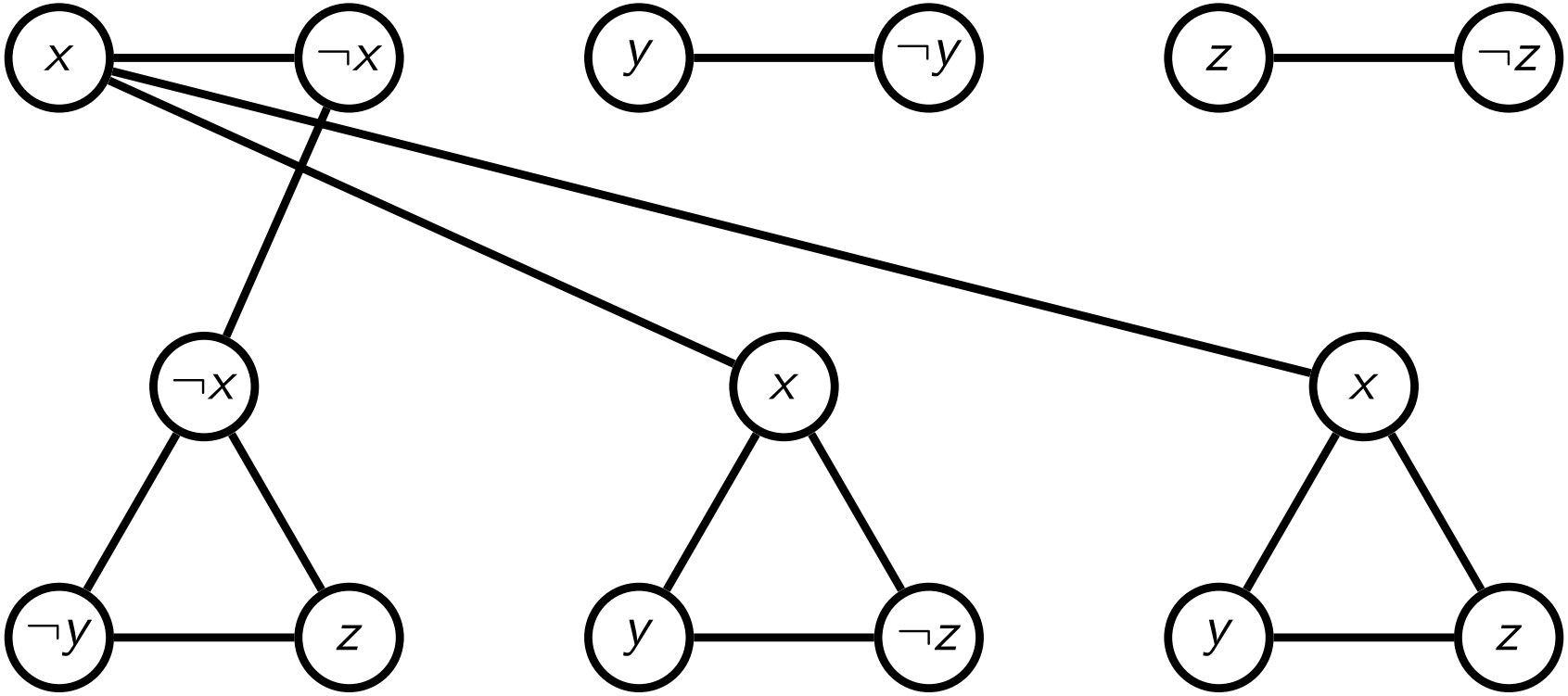

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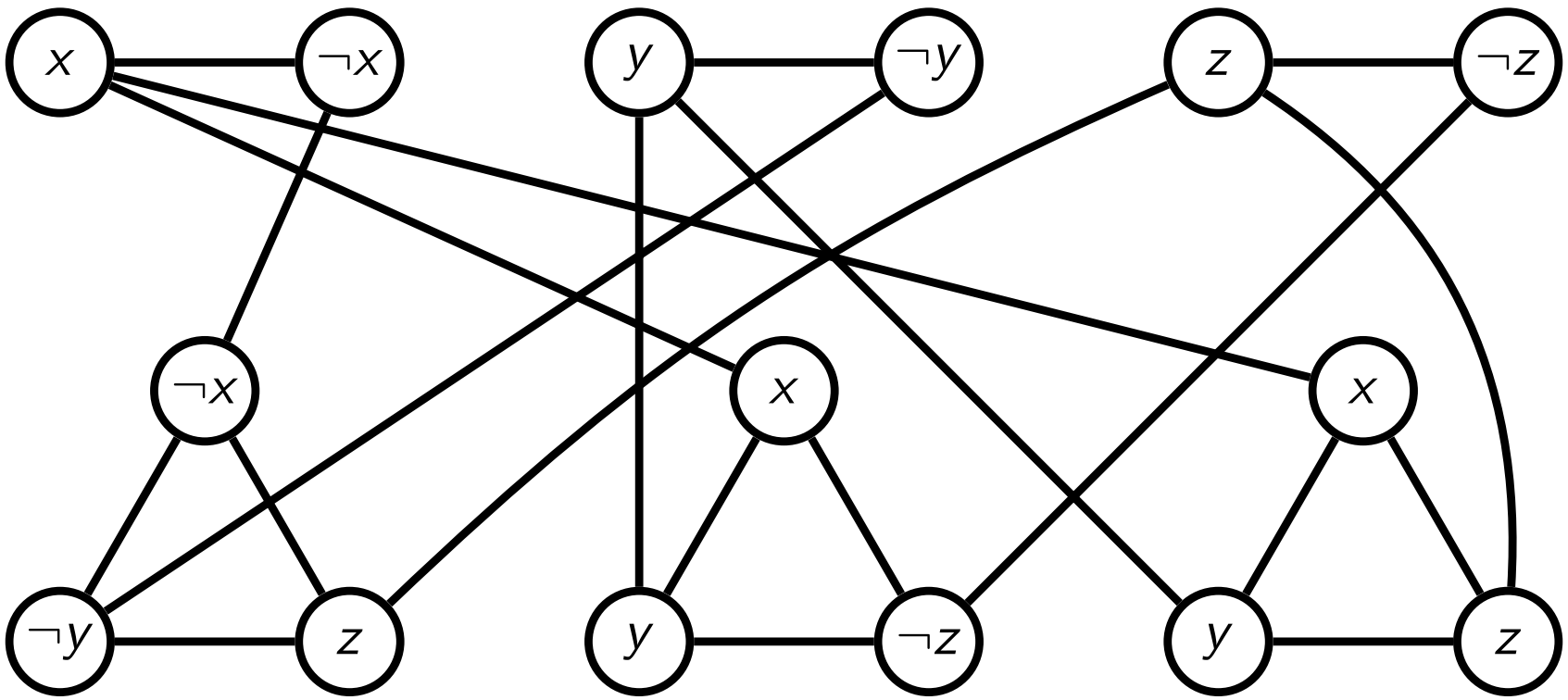

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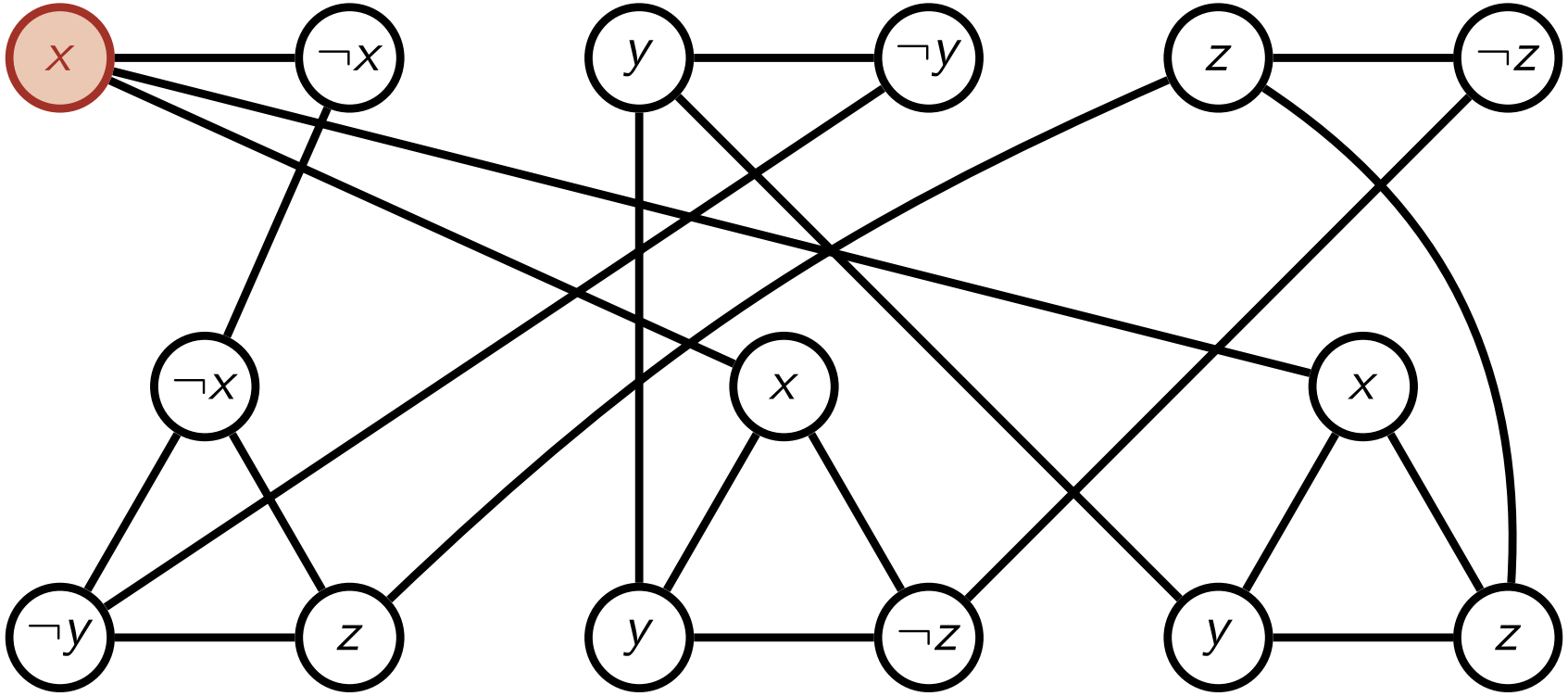
- The resulting graph has a $v + 2c = 9$-vertex cover iff φ is satisfiable.



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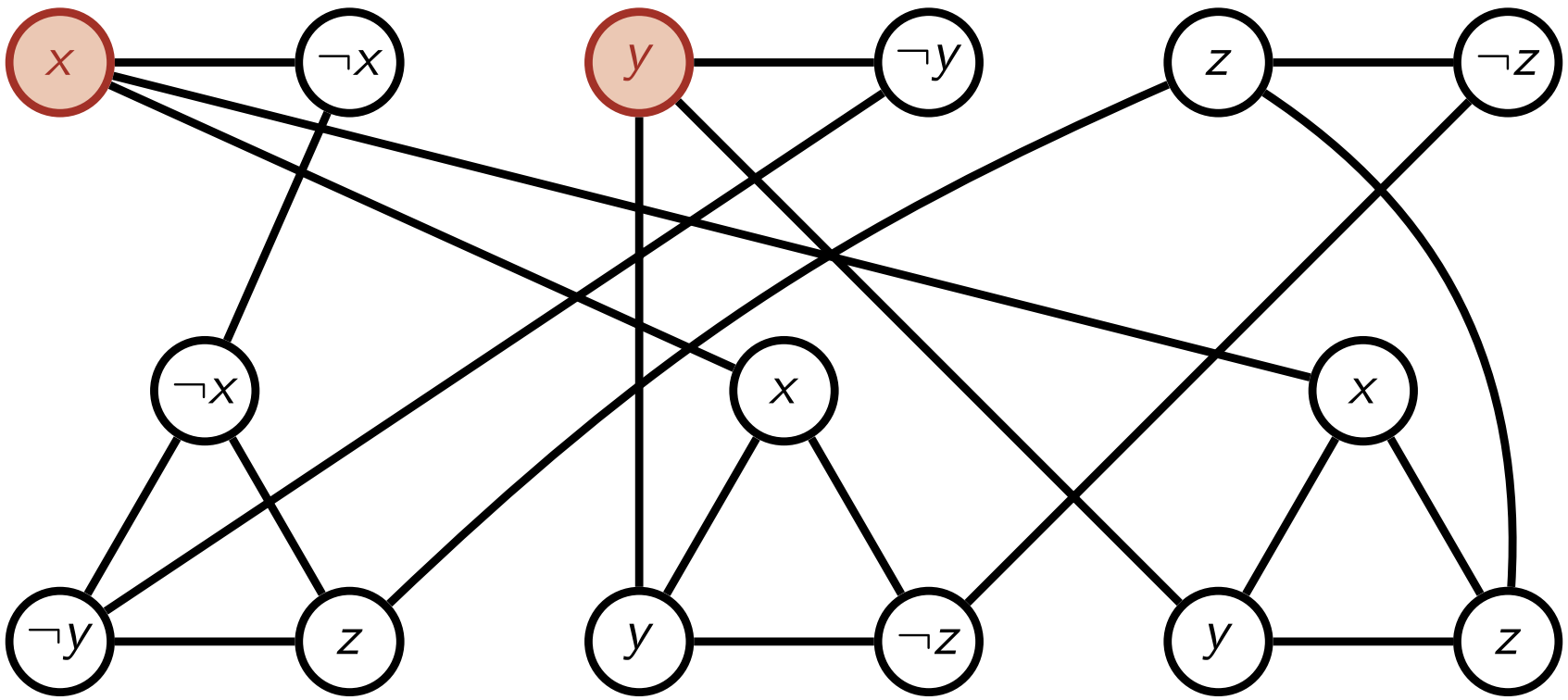

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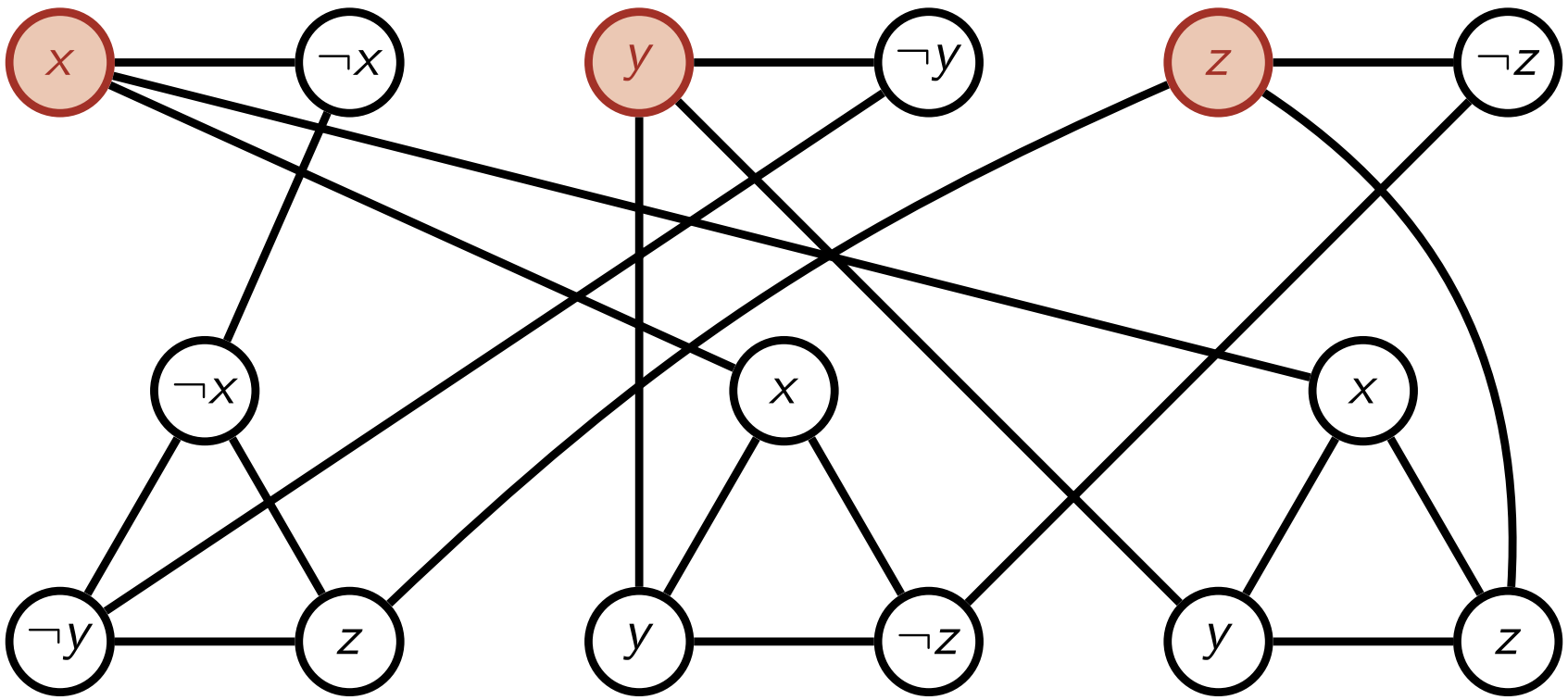

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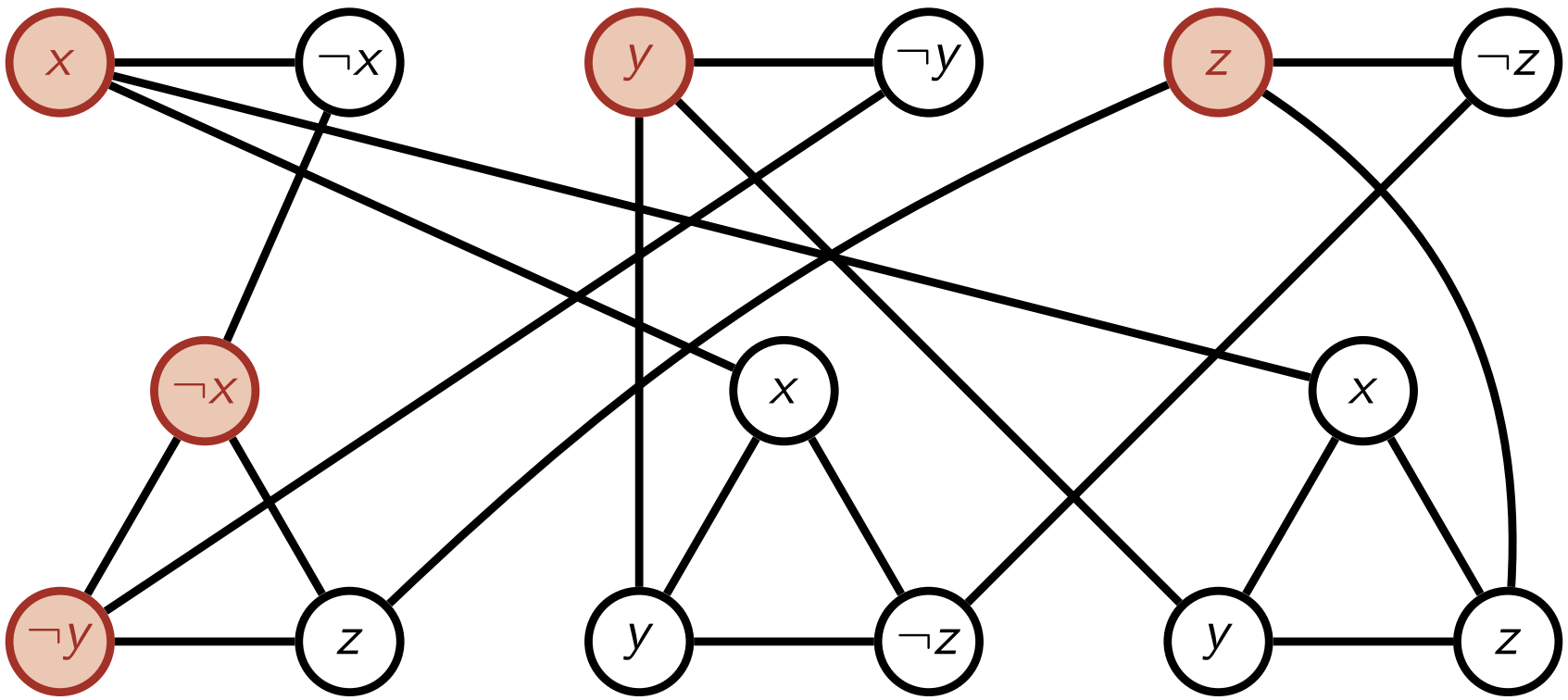

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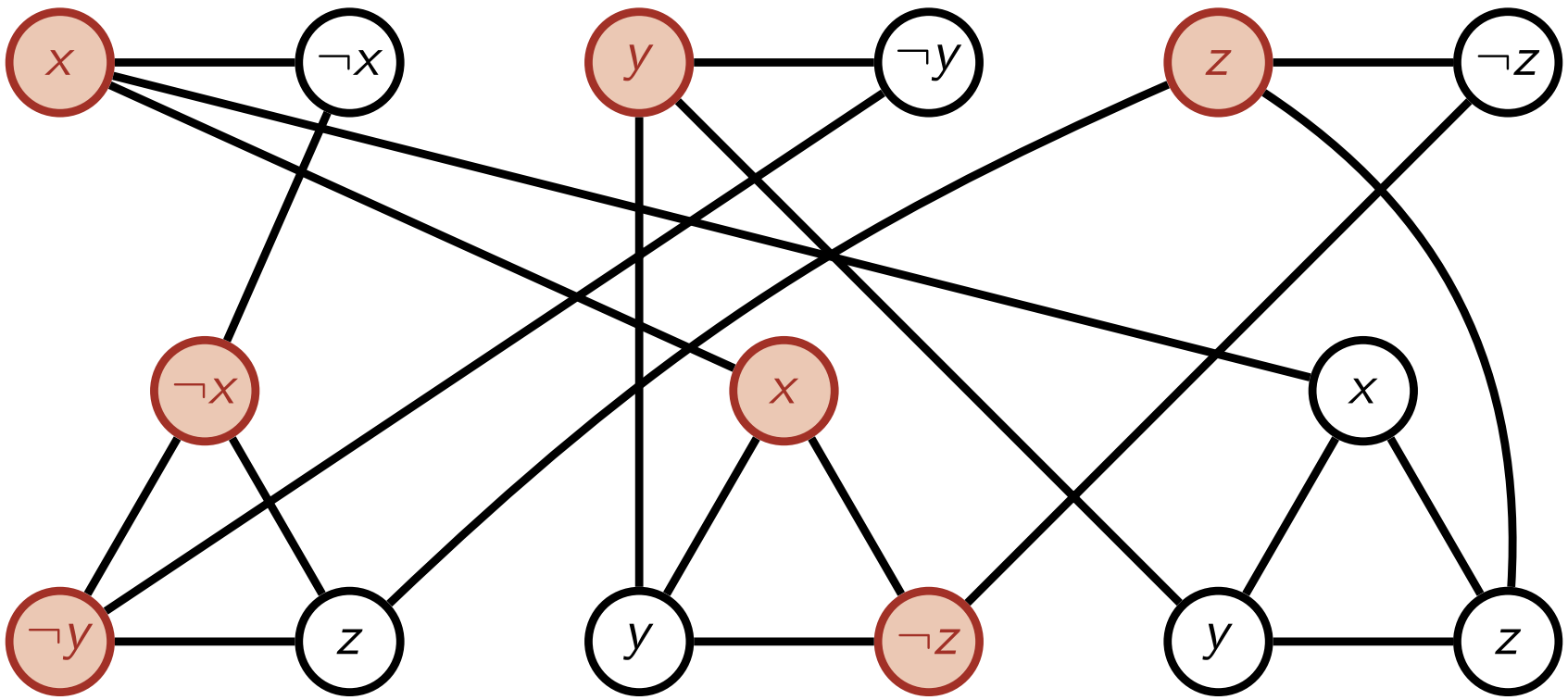

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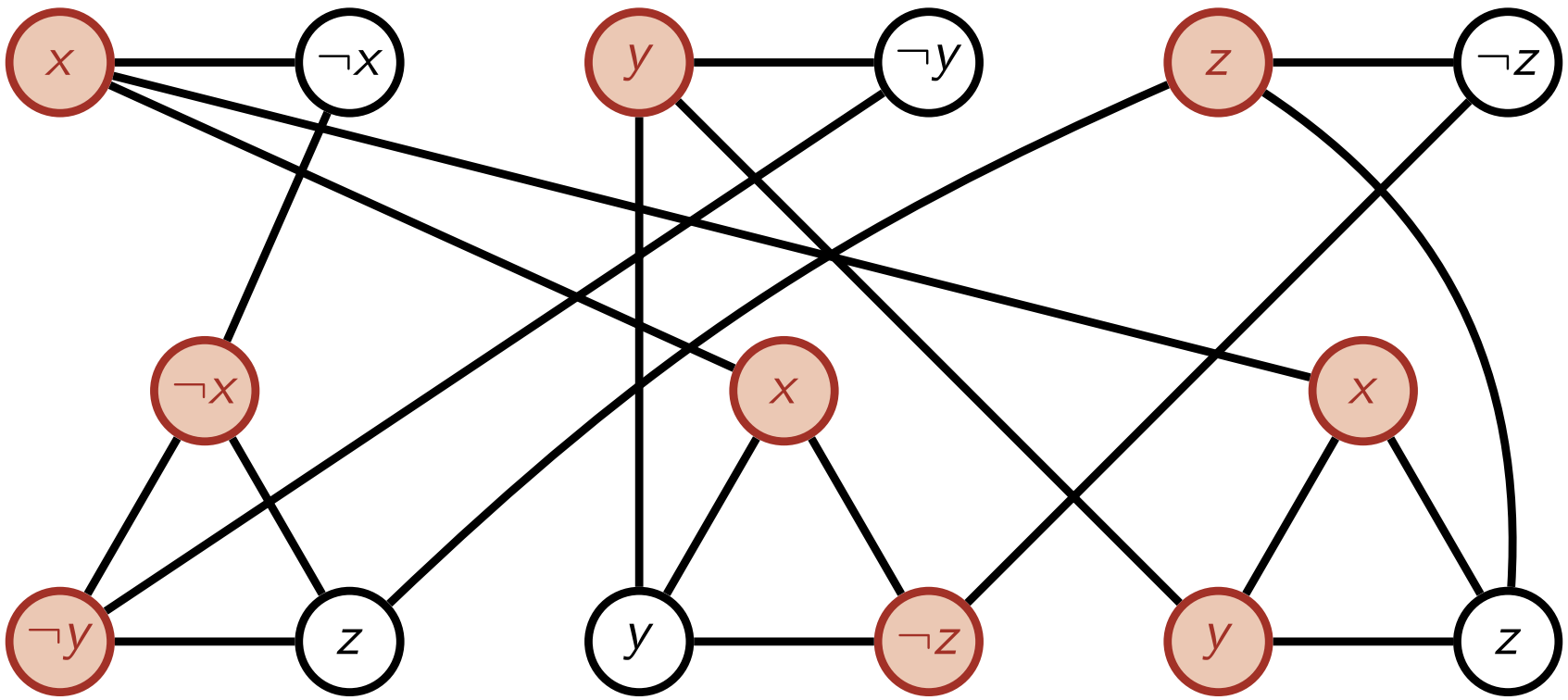

There is Much More

Many more problems are NP-complete.

- Karp '72: 21 NP-complete problems, among them SAT, $3\mathrm{SAT}$, CLIQUE, VC, Hamiltonian path...
- Garey & Johnson '79: a book with 300 NP-hard problems.
- More modern list: `https://en.wikipedia.org/wiki/List_of_NP-complete_problems`

Conclusion

We have seen

- the hardest problems in NP, the NP-complete ones.
- Many useful problems are NP-complete.
- If we can find a polynomial-time algorithm for one such problem, then we have one for all problems in NP, i.e., $\mathrm{P} = \mathrm{NP}$.
- Researchers have for more than 60 years, without success, tried to resolve the P vs. NP.
- Note: Public-key cryptography depends on the assumption that some problems cannot be solved efficiently!

Reading

- Unfortunately, our book uses a completely different approach to NP-completeness.
- Nevertheless, read Section 3.5 on NP-completeness with that in mind.